Hartwig Medical Foundation

oncoanalyser
ORANGE Report v3.0

User Manual

More technical information can be found here: https://github.com/hartwigmedical/hmftools

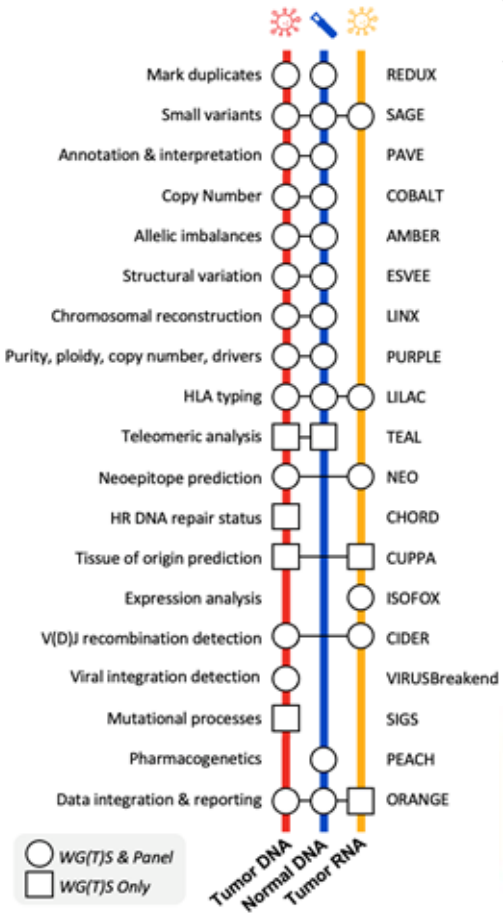
Contents

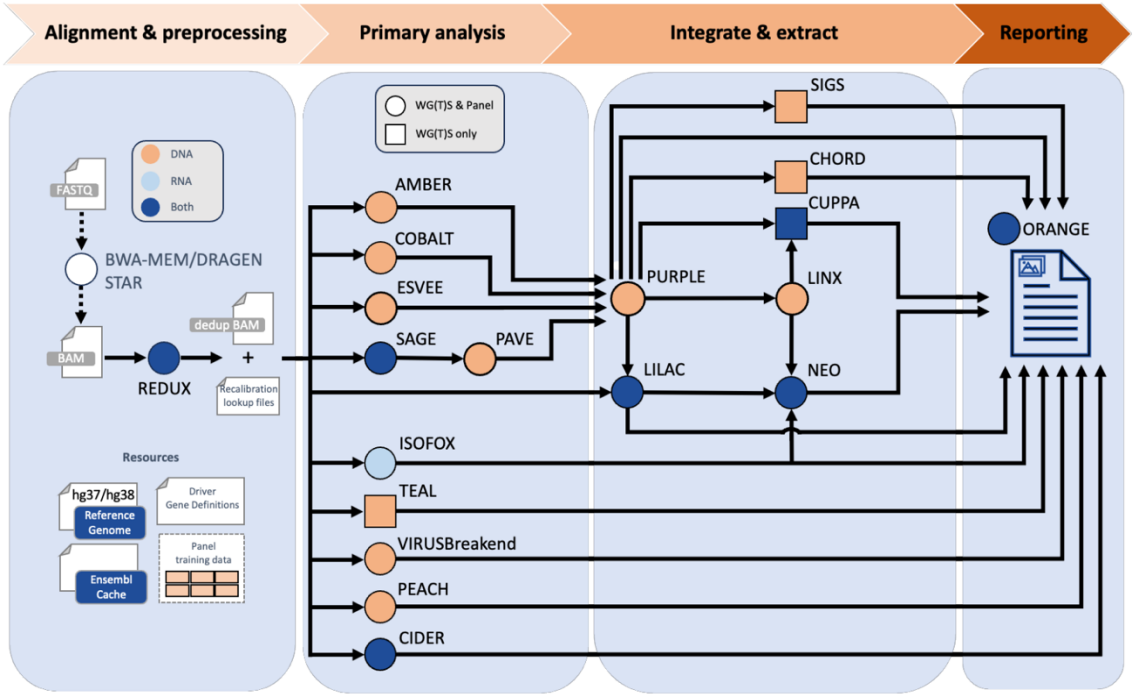
1. Report overview & conventions
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4. Somatic variants
5. Germline variants
6. RNA
7. Immunology
8. Tissue of origin
9. QC
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Report Overview & Conventions

Oncoanalyser Pipeline of tools used to generate an Orange Report





SAMPLE SOURCE

- Fresh-frozen tissue
- FFPE
- Liquid biopsy (plasma, CSF, ascites, pleural fluid)

MODUS

- Tumor-normal pairs
- Tumor only
- Longitudinal samples
- With and without RNA-seq

APPLICATION

- Whole Genome Sequencing
- Whole Exome Sequencing
- Large panel
- Biomodal

SEQUENCING PLATFORM

- Illumina
- Ultima Genomics
- Roche SBX

Oncoanalyser Orange Report Intent

The Orange Report provides a **comprehensive summary** of oncoanalyser outputs and is automatically issued for every sample analysed (both in the free oncoanalyser version avaliable from nf-core or the OncoAnalyser Dx SaaS solution)

The report lists **clinically relevant variants and biomarkers** found in a sample in a semi-annotated way and is optimised for the input type (t/n; tumor only, DNA/RNA, FFPE/FF) and analysis performed (WGS/specified panel)

It uses Hartwig's default driver logic and latest oncoanalyser version (unless customised)

It is a non-curated report and is **NOT suitable for use as a final clinical patient report**

Intended users include researchers, clinical curation teams or Molecular Oncology Boards that want to review detailed information about all molecular characteristics of a tumor

If you wish to curate Orange reports or customise driver logic please use the OncoAnalyser Dx SaaS platform

Report structure

The Orange Report consists of 8 subsections

Section	Contents
Summary Page	• Key drivers & genome wide biomarkers • Overall CIRCOS picture summarising molecular events
Somatic Variants	• Somatic small variants, copy number alterations, fusions, disruptions, viruses • Somatic molecular signatures • Plots of structural variant associated somatic drivers.
Germline Variants	• Germline small variants, copy number alterations, disruptions & chromosomal aberrations • Pharmacogenetics
RNA	• High expression • Fusions and novel splice junctions not detected in DNA
Immunology	• HLA Class I and Class II alleles
Tissue of Origin	• Comprehensive report showing CUPPA classifiers
Quality Control (QC)	• Comprehensive report including key metrics and plots of various QC quantities
Purity & Ploidy	• Plots showing purity and ploidy inputs and QC reports

Configuration of the Orange report which drives automation

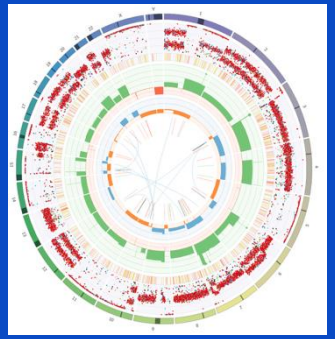
The content of the Orange report is configured in the driver gene panel. The driver gene panel contains a list of genes of interest and a set of flags to control which variant types should be reported for that gene.

Each gene is also configured as an “ONCO” or “TSG” which impacts the way the algorithm for ‘driver likelihood’ is calculated to prioritise small variants. If TSG is configured, the biallelic status is taken into account and genes with multiple mutations will have additive likelihoods. Biallelic nonsense & canonical splice variants are given a likelihood of 1. For “ONCO” types, in-frame indels outside of repeat contexts are given a likelihood of 1. Known pathogenic sites for both types are also given likelihood of 1.

gene	reportMissense	reportNonsense	reportSplice	reportDeletion	reportHetDeletion	reportLOH	reportDisruption	reportAmplification	AmplificationRatio	reportSomaticHotspot	likelihoodType	reportGermlineVariant	reportGermlineHotspot	reportGermlineDisruption	reportGermlineDeletion	additionalReportedTranscripts	reportPGX
CCND1	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	3	TRUE	ONCO	NONE	NONE	NONE	NONE		FALSE
CDKN2A	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	3	TRUE	TSG	ANY	ANY	ANY	ANY	ENST00000579755	FALSE
DPYD	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	3	FALSE	TSG	NONE	NONE	NONE	NONE		TRUE
EGFR	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	3	TRUE	ONCO	VARIANT_NOT_LOST	VARIANT_NOT_LOST	NONE	NONE		FALSE
PTEN	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	3	TRUE	TSG	ANY	ANY	ANY	ANY		FALSE
TERT	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	3	TRUE	ONCO	NONE	ANY	NONE	NONE		FALSE
TP53	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	3	TRUE	TSG	ANY	ANY	ANY	ANY		FALSE

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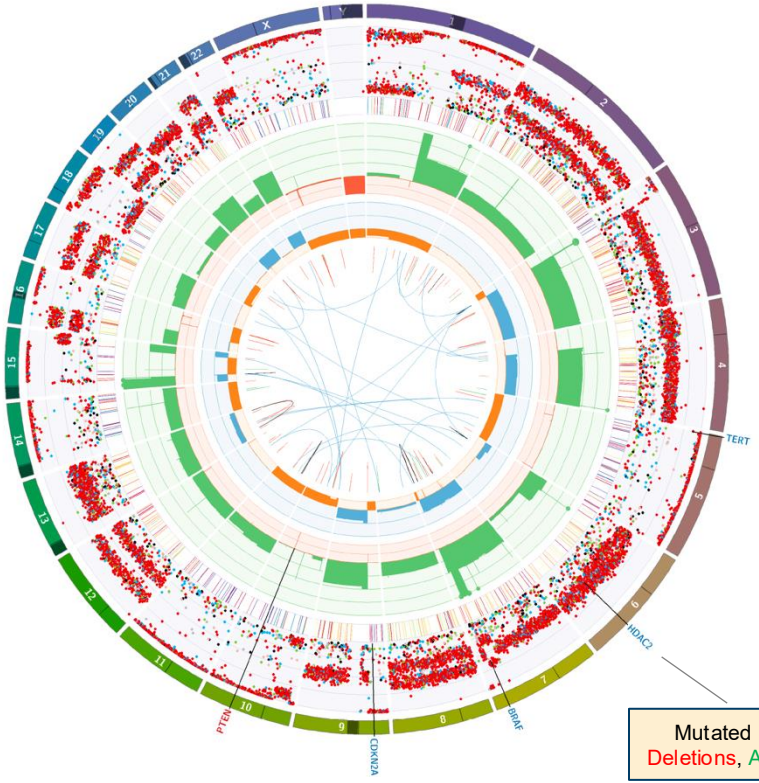
Guide to CIRCOS Plots



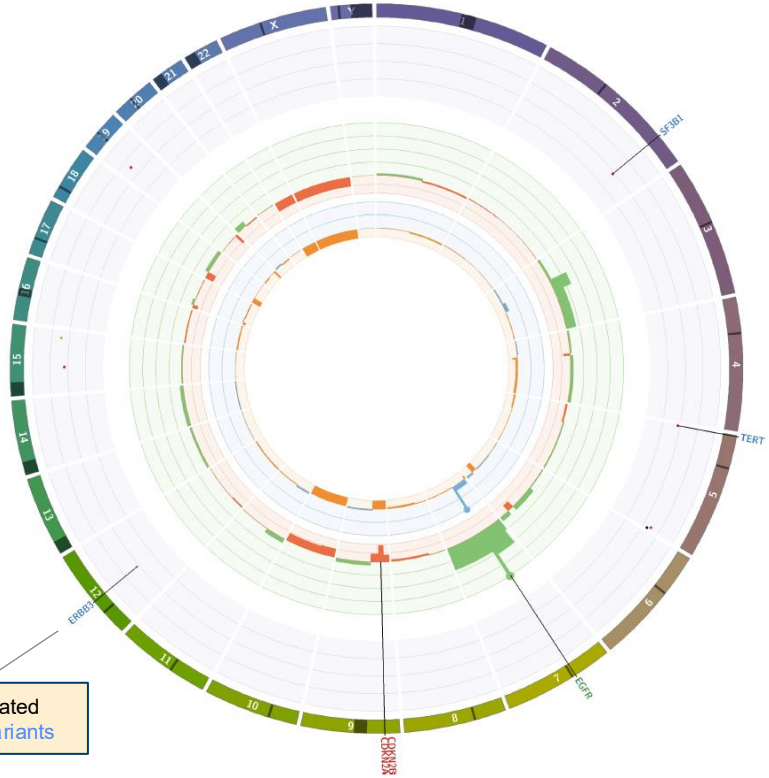


CIRCOS plot – a visual summary of tumor molecular events

Whole Genome Sequencing CIRCOS Plot

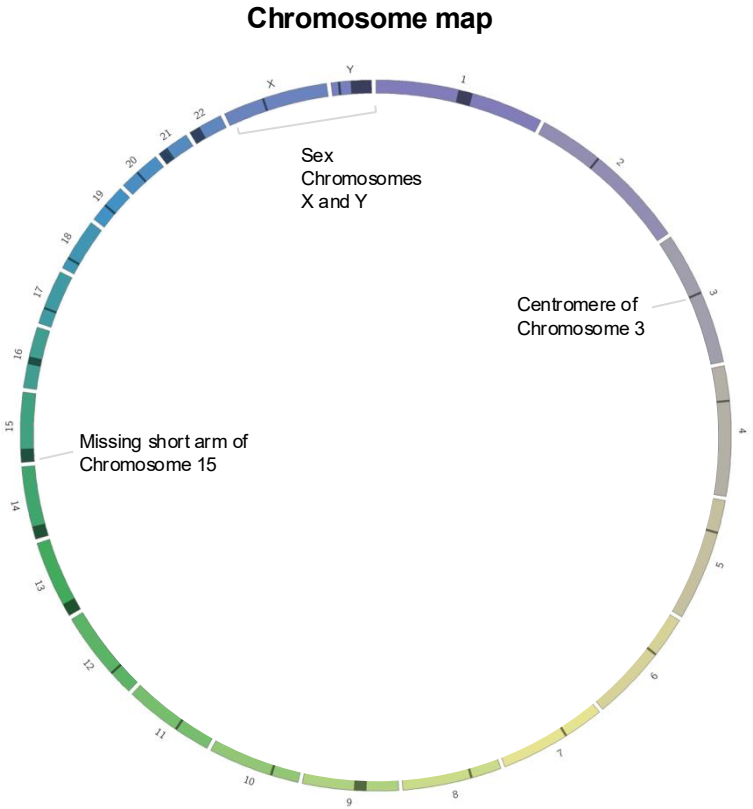


Panel (TSO500) CIRCOS Plot



Mutated driver genes are annotated
Deletions, Amplifications, Small Variants

Understanding CIRCOS plots - Chromosomes



All 22 chromosomes are numbered, plus X and Y, and are drawn to scale.

Darker shaded areas represent gaps in the human reference genome eg centromeres, heterochromatin & missing short arms.

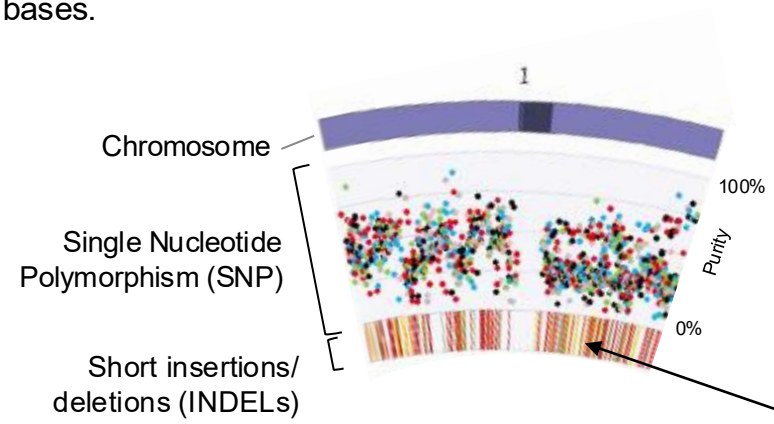
Understanding CIRCOS plot – SNPs and INDELS

The next layers show all tumor observed variants (incl. exon, intron and intergenic regions) and are divided into an outer ring of single nucleotide polymorphism (SNP) allele frequencies and an inner ring of short insertion/deletion (INDEL) locations of < 50 bases.

Each dot represents a single variant and are coloured according to the type of base change as described in Alexandrov et al. 2013 Nature.

- C>A
- C>G
- C>T
- T>A
- T>C
- T>G

Melanomas with UV signatures will normally appear dominated by red dots (C>T) whereas Lung cancers with smoking signatures will predominantly have blue dots (C>A)



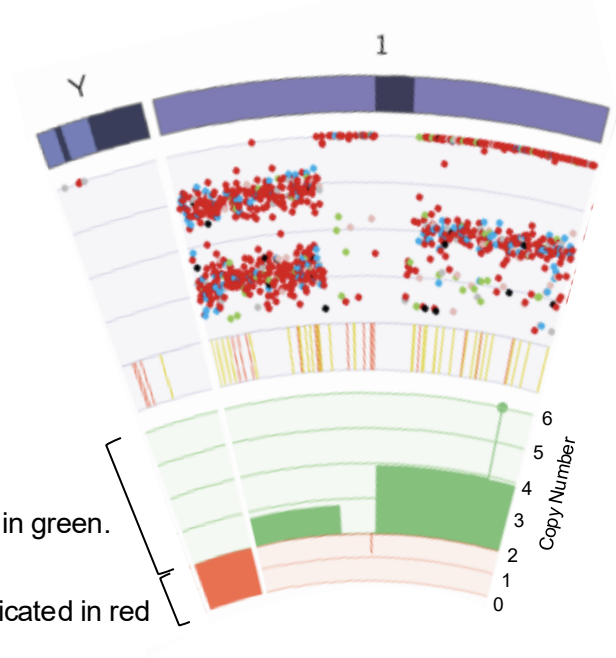
Variant allele frequencies have been corrected for tumor purity and scale from 0 to100%.

The high density of INDELs in this example suggests a likely MSI sample

INDELs are colored:
- yellow for insertions
- red for deletions
- green for deletions with microhomology

Understanding CIRCOS plots – Copy Number

The next layers shows all observed tumor purity adjusted copy number changes, including both focal and chromosomal events.



Copy number gains are shown in green.

Copy number losses are indicated in red

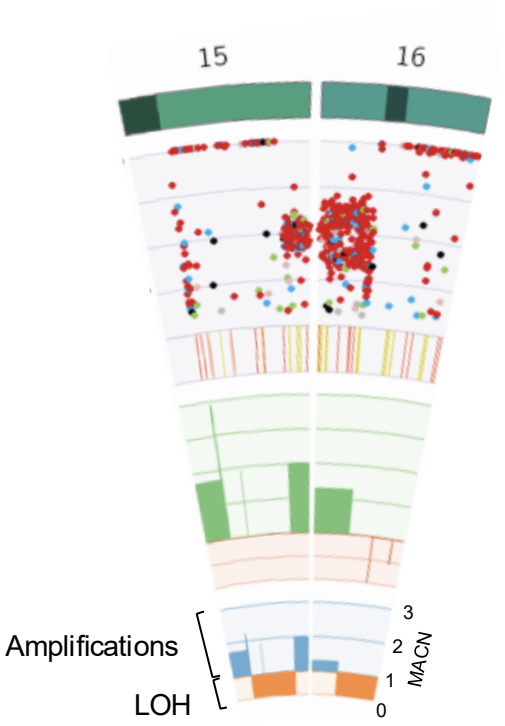
The copy number scale ranges from 0 (complete loss) to 6 (high level gains).

If the absolute copy number is > 6 it is shown as 6 with a green dot on the diagram (example on chr. 1q).

Homozygous loss is normally highly localised in cancer with the loss of Y chromosome shown in this picture a notable exception

As a diploid organism we expect 2 copies of every autosome as baseline in normal tissue

Understanding CIRCOS plots – Minor Allele Copy Number



The next layer represents the observed 'minor allele copy numbers' (MACN) across the chromosome.

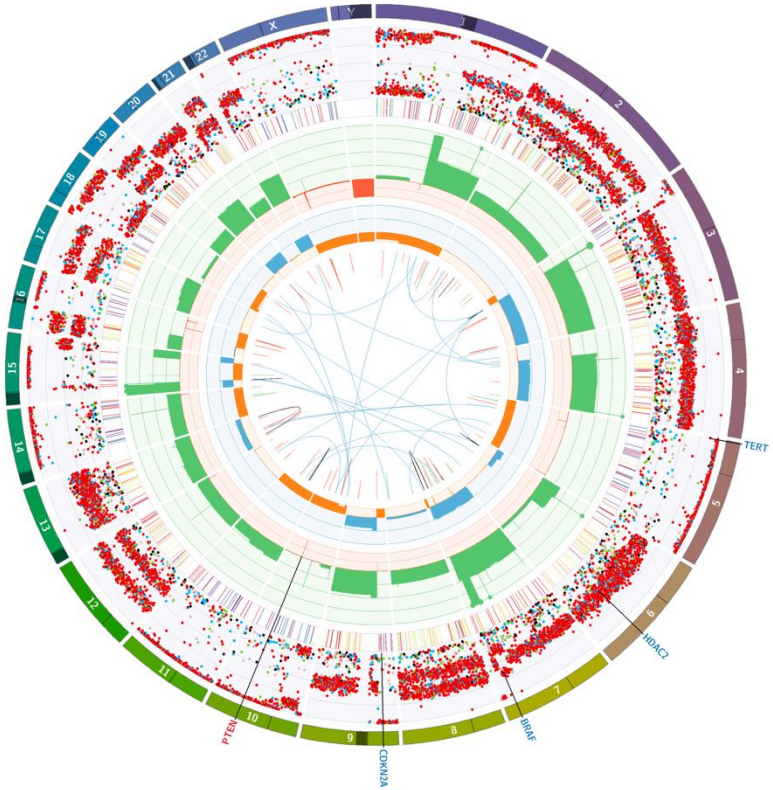
Minor allele copy number represents the specific copy count of the less frequent (or lost) allele at a given locus, distinguishing between different allelic configurations that total copy number analysis misses.

It is crucial for identifying allelic imbalances, Loss of Heterozygosity (LOH), and tumor-specific gains.

The range of the chart is from 0 to 3.
The expected normal minor allele copy number (MACN) is 1.
Anything below 1 is shown as a loss and represents an LOH event (orange).
Minor allele copy numbers above 1 indicate amplification events of both A and B alleles at the indicated locations (blue).

Understanding CIRCOS Plots – Structural Variants

WGS CIRCOS plot

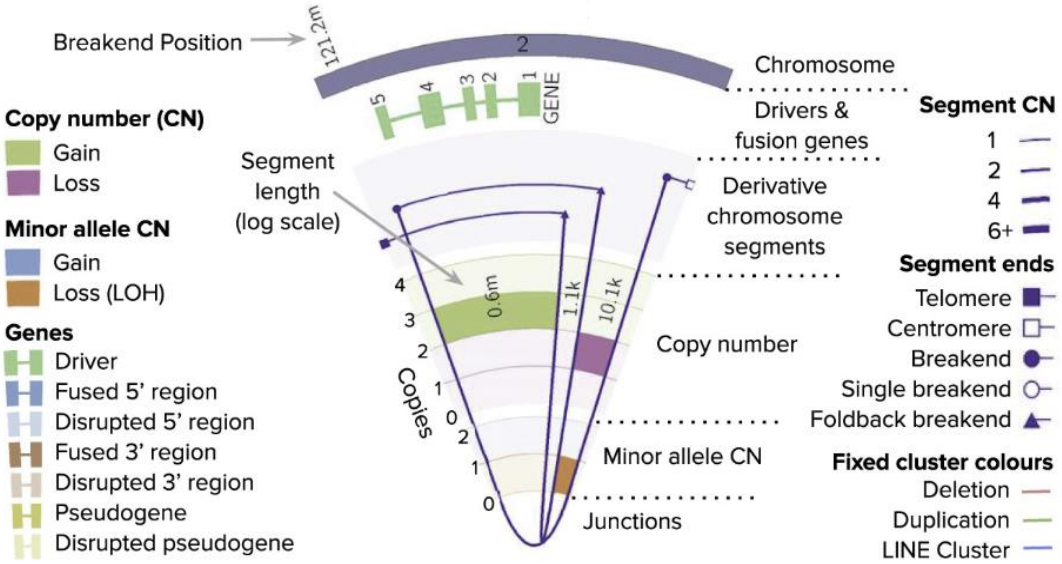


The innermost circle displays the observed structural variants within or between the chromosomes.

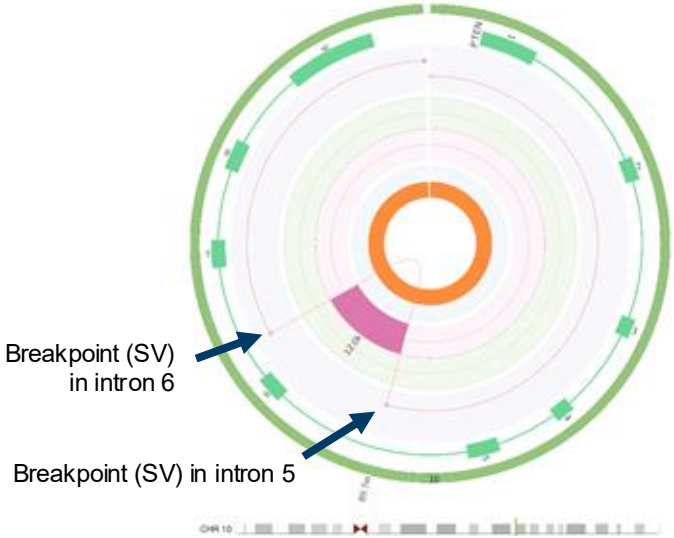
Translocations are indicated in **blue**
deletions in **red**
insertions in **yellow**
tandem duplications in **green**
inversions in **black**

For local chromosomal alterations the length of the line is proportional to the log of the length of the variant

Structural Variant CIRCOS - LINX Visualisation Guide



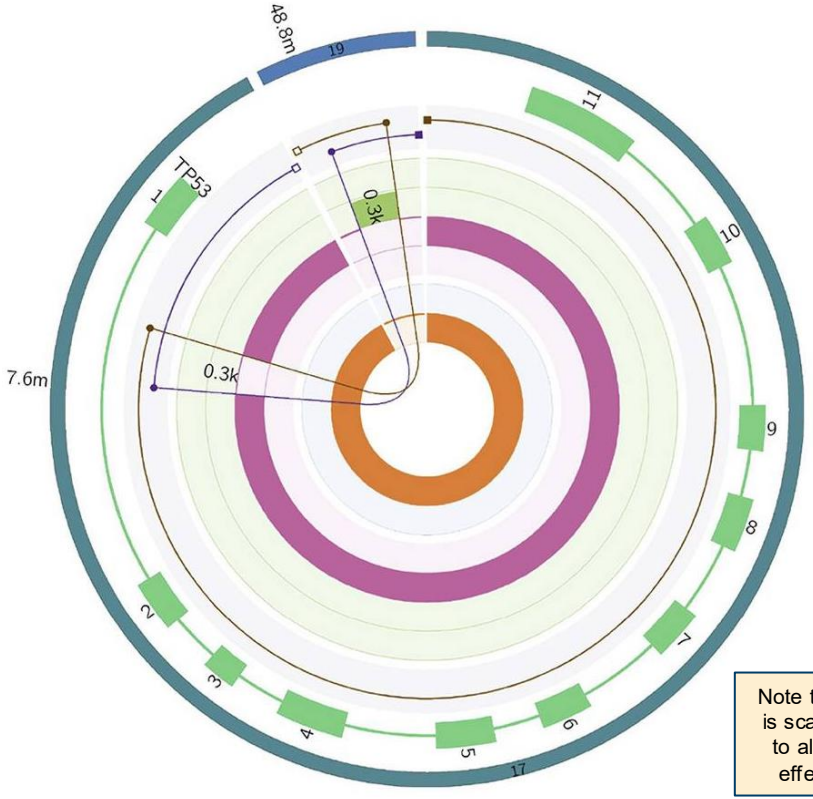
Chromosome mode: All clusters touching chromosome shown. Each cluster has a different colour
Cluster mode: Only variants in cluster shown. Each derivative chromosome has a different colour



Example of a partial homozygous deletion of PTEN exon 6 on chromosome 10

Shale et al (2022) Cell Genome

LINX Visualisation – example homozygous disruption



Reciprocal translocation affecting *TP53* in a prostate tumor.

The two predicted derivative chromosomes overlap by approximately 300 bases on both ends but are *trans* phased, which rules out the possibility of a templated insertion at either location.

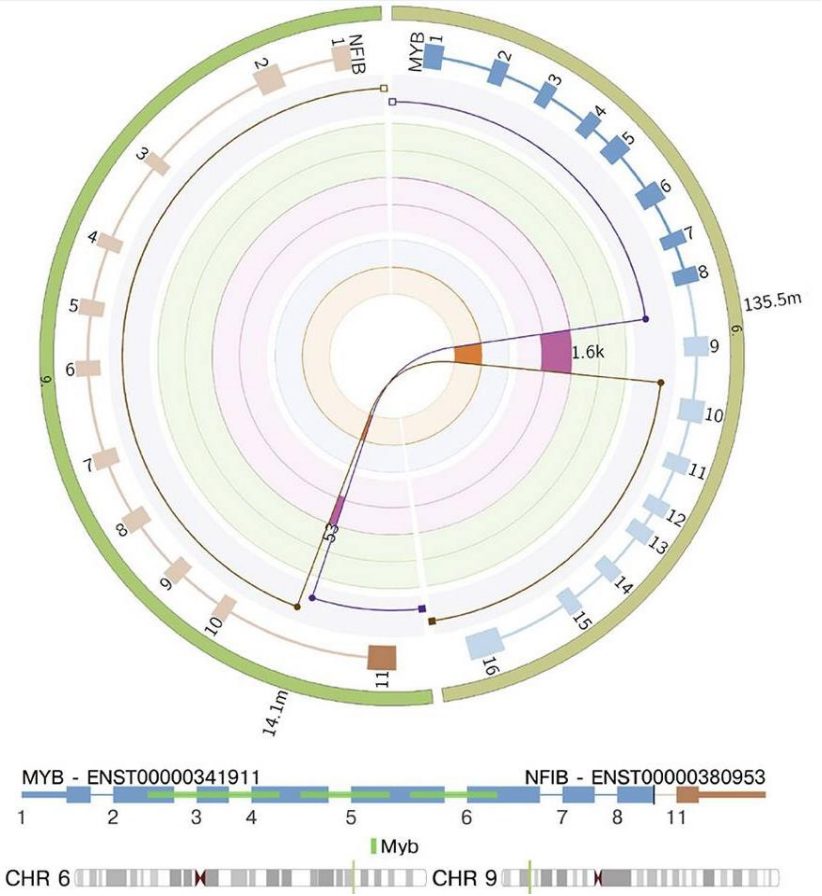
Although the *TP53* copy number alternates between one and two, no derivative chromosome contains the full gene and the gene is homozygously disrupted.

Note that the size of each feature is scaled by the log of the length to allow small- and large-scale effects to be viewed together



Shale et al (2022) Cell Genome

LINX Visualisation – example fusion



A *MYB-NFIB* fusion caused by a reciprocal translocation in a salivary gland tumor.

The translocation links exons 1–8 in *MYB* to exon 11 in *NFIB* (purple line)

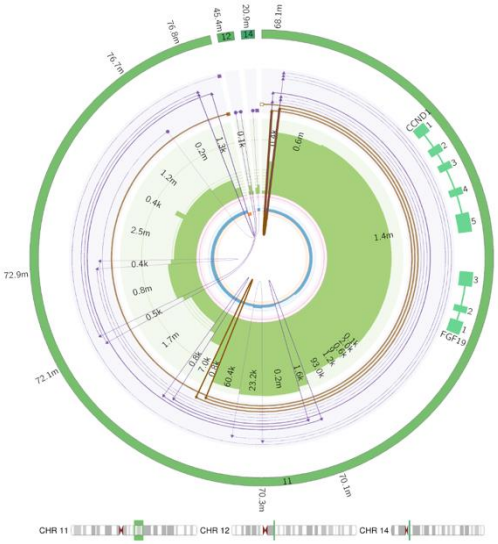
Shale et al (2022) Cell Genome

Linx Visualisation – example amplifications & deletions

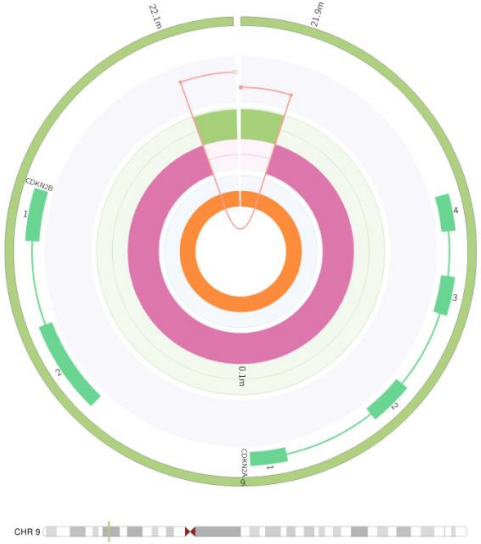
- **Full gain**: Copy number (CN) ≥ 3 * ploidy
- **Partial gain**: only a part of the gene has CN ≥ 3 * ploidy
- **Full loss**: CN < 0.5
- **Partial loss**: only a part of the gene has CN < 0.5

Visualization examples (LINX plots)

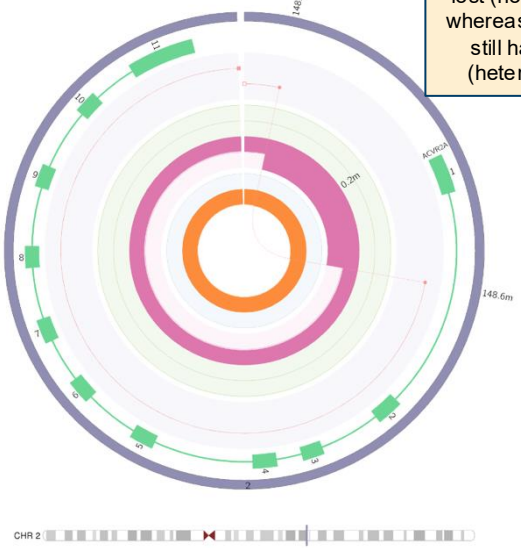
Full gain of CCND1



Full loss of CDKN2A



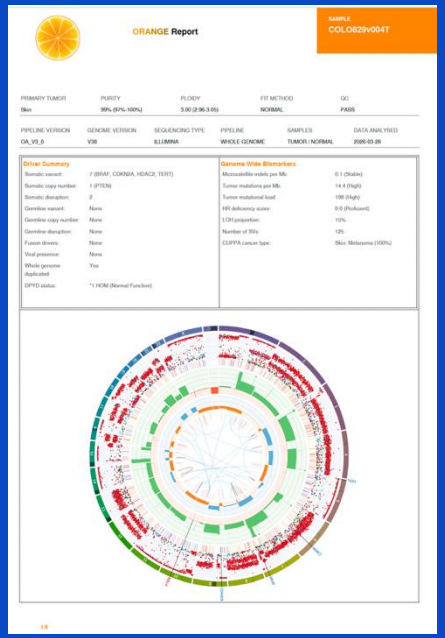
Partial loss of AVCR2A



Only exon 1 is completely lost (homozygous deletion), whereas the rest of the gene still has a wildtype copy (heterozygous deletion)

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Summary Page



ORANGE Report

SAMPLE
COLO829v004T

PRIMARY TUMOR	PURITY	PLOIDY	FIT METHOD	QC
Skin	95% (87%-100%)	3.00 (2.96-3.05)	NORMAL	PASS

PIPELINE VERSION	GENOME VERSION	SEQUENCING TYPE	PIPELINE	SAMPLES	DATA ANALYSED
OA_V3_0	V38	ILLUMINA	WHOLE GENOME	TUMOR / NORMAL	2020-03-26

Driver Summary

Somatic variant:	7 (BRAF, CDKN2A, HDAC2, TERT)
Somatic copy number:	1 (PTEN)
Somatic disruption:	2
Germline variant:	None
Germline copy number:	None
Germline disruption:	None
Fusion drivers:	None
Viral presence:	None
Whole genome duplicated:	Yes
DPYD status:	*1 HOM (Normal Function)

Genome Wide Biomarkers

Microsatellite indels per Mb:	0.1 (Stable)
Tumor mutations per Mb:	14.4 (High)
Tumor mutational load:	100 (High)
HR deficiency score:	0.0 (Proficient)
LOH proportion:	11%
Number of SVs:	125
CUPPA cancer type:	Skin: Melanoma (100%)

Summary Page – Sample Details



ORANGE Report

SAMPLE
COLO829v004T

PRIMARY TUMOR	PURITY	PLOIDY	FIT METHOD	QC
Skin	99% (97%-100%)	3.00 (2.96-3.05)	NORMAL	PASS

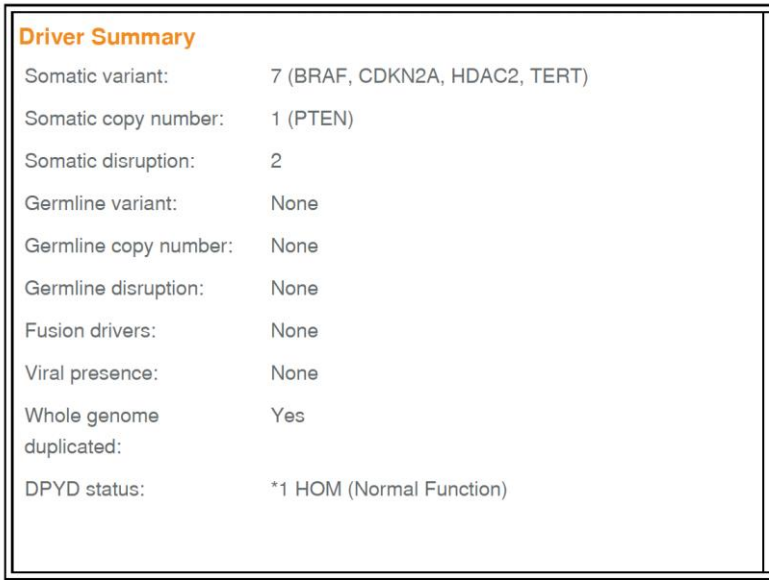
PIPELINE VERSION	GENOME VERSION	SEQUENCING TYPE	PIPELINE	SAMPLES	DATA ANALYSED
OA_V3_0	V38	ILLUMINA	WHOLE GENOME	TUMOR / NORMAL	2026-03-26

Field	Description
Primary tumor	Tumor type as provided by the hospital
Purity	% of tumor cells in the tumor sample
Ploidy	Average number of chromosome copies in the tumor
Fit method	Method used to fit purity & ploidy. Either 'NORMAL' (based on aneuploidy), 'SOMATIC' (using somatic small variants) or 'NO_TUMOR'
QC	Either PASS or will specify one or more WARN or FAIL reason (see table)
Pipeline version	onconaliser version used to generate analysis and report
Sequencing type	Sequencing machine type for data analysed. One of {ILLUMINA, ULTIMA, SBX}
Pipeline	May be either 'WHOLE GENOME' or 'TARGETED'
Samples	Specifies combination of TUMOR, NORMAL & RNA run
Date analysed	Date report generated

QC warning and fail statuses

WARN / FAIL Status	Meaning
WARN_DELETED_GENES	>280 genes are deleted genome wide
WARN_LOW_PURITY	Purity < 20%
WARN_TINC	Tumor in normal contamination > 0.5%
WARN_HIGH_COPY_NUMBER_NOISE	Sample has > 220 CN segments unsupported by SV
WARN_GENDER_MISMATCH	Gender from chrY CN <> Gender from chrX BAF
FAIL_TINC	Tumor in normal contamination > 25%
FAIL_CONTAMINATION	Contamination (human) > 10% [can indicate chimerism]
FAIL_NO_TUMOR	No evidence of tumor purity > 8%

Summary Page – Driver Summary



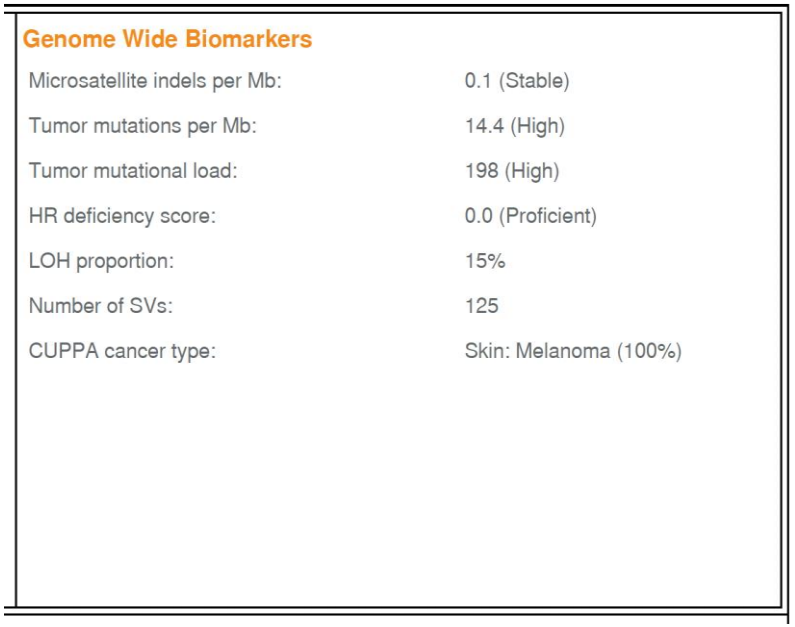
Driver Summary

Somatic variant:	7 (BRAF, CDKN2A, HDAC2, TERT)
Somatic copy number:	1 (PTEN)
Somatic disruption:	2
Germline variant:	None
Germline copy number:	None
Germline disruption:	None
Fusion drivers:	None
Viral presence:	None
Whole genome duplicated:	Yes
DPYD status:	*1 HOM (Normal Function)

Field	Description
Somatic variant	# of reportable small variants. Somatic variants shown in brackets have driver likelihood > 20%
Somatic copy number	# of reportable amplifications & deletions. Shown in brackets if high driver likelihood, ie if homozgous deletion [minCN < 0.5] or if amplification exceeds N * ploidy [default=3]
Germline variant	# of reportable germline small variants
Germline copy number	# of reportable germline amplifications & disruptions
Germline disruption	# of reportable germline disruptions.
Fusion drivers	# of reportable pathogenic gene fusions.
Viral presence	List of viruses detected in the tumor sample
Whole genome duplicated	Yes = whole genome doubling event detected. No = not detected
DPYD status	Allele and zygosity of DPYD for pharmacogenetics purposes

Summary page – Genome wide biomarkers

Field	Description
Microsatellite indels per Mb	# of indels in repeats in microsatellites across the genome per Mb. Unstable if > 4
Tumor mutations per Mb	# of all somatic variants across the genome per Mb. High if TMB>=10, low if TMB <10
Tumor mutational load	# of all somatic missense mutations across the genome. High if TML>=140, low if TMB <140
HR deficiency score	Likelihood of HR deficiency as assessed by CHORD based off molecular profile. Proficient if < 0.5 else deficient
LOH proportion	Percentage of autosomes with loss of heterozygosity genome wide
Number of SVs	Total # of SVs found in sample
CUPPA cancer type	Cancer type predicted by CUPPA (tissue of origin algorithm). Confidence shown in brackets. See detailed report section for details



Genome Wide Biomarkers

Microsatellite indels per Mb:	0.1 (Stable)
Tumor mutations per Mb:	14.4 (High)
Tumor mutational load:	198 (High)
HR deficiency score:	0.0 (Proficient)
LOH proportion:	15%
Number of SVs:	125
CUPPA cancer type:	Skin: Melanoma (100%)

Summary page - Algorithm details

Microsatellite Indels Per Mb

- Microsatellites are regions of short repetitive DNA. DNA mismatch repair (MMR) system (MLH1, MSH2, MSH6 and PMS2) detects and corrects replication errors. If MMR is not working properly, mutations will accumulate in the cell
- Microsatellite status: represents the number of somatic inserts and deletes in microsatellites per Mb. ≥ 4 is considered microsatellite instable (MSI). <4 is microsatellite stable (MSS)
- For WGS the rate of microsatellite indels is counted directly. For targeted sequencing it is estimated via observing the error rate in microsatellites across all panel regions for long homopolymers and dinucleotide repeats
- Microsatellite status is an important predictive biomarker for immunotherapy response

HR Deficiency

- Homologous recombination (HR) is an essential pathway for DNA double-strand break repair
- Mutations in essential HR genes, including BRCA1 and BRCA2, may result in HR deficiency (HRD)
- Oncoanalyser predicts a HRD score between 0 & 1. >0.5 is HRD+
- For WGS, a logistic regression classifier, CHORD, is used to predict HRD based on associated molecular indel & SV signatures
- For targeted analyses, an image-based classifier, visual CHORD (vCHORD), can be trained per panel to also predict HRD

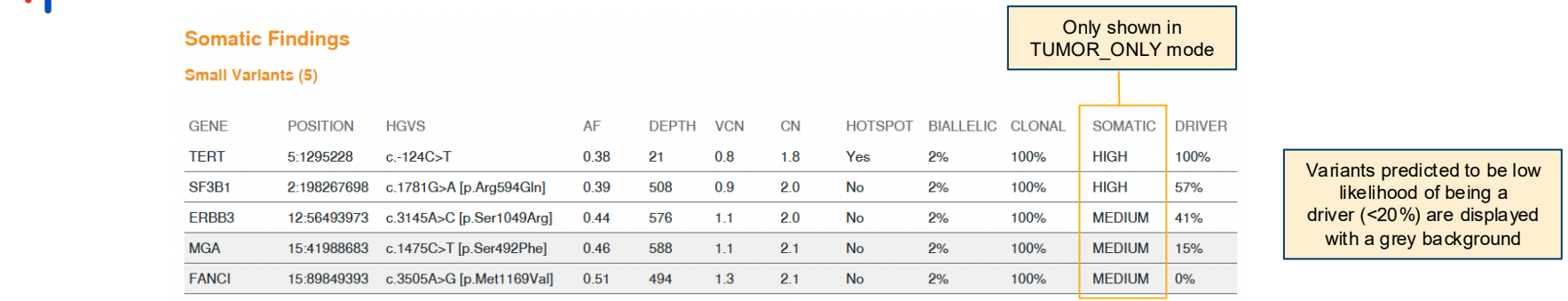
TMB / TML

- Tumor mutational burden (TMB) is the density of all small variant mutations per Mb
- Tumor mutational load (TML) is the count of missense mutations in the tumor genome wide
- For WGS, oncoanalyser counts TMB & TML directly
- For targeted analyses, TMB & TML are estimated based on the rate of likely-somatic variants in the coding on-target regions
- Both TMB & TML are predicitive biomarkers for immunotherapy

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Somatic Variants

Somatic Variants – Small Variants



Somatic Findings

Small Variants (5)

GENE	POSITION	HGVS	AF	DEPTH	VCN	CN	HOTSPOT	BIALLELIC	CLONAL	SOMATIC	DRIVER
TERT	5:1295228	c.-124C>T	0.38	21	0.8	1.8	Yes	2%	100%	HIGH	100%
SF3B1	2:198267698	c.1781G>A [p.Arg594Gln]	0.39	508	0.9	2.0	No	2%	100%	HIGH	57%
ERBB3	12:56493973	c.3145A>C [p.Ser1049Arg]	0.44	576	1.1	2.0	No	2%	100%	MEDIUM	41%
MGA	15:41988683	c.1475C>T [p.Ser492Phe]	0.46	588	1.1	2.1	No	2%	100%	MEDIUM	15%
FANCI	15:89849393	c.3505A>G [p.Met1169Val]	0.51	494	1.3	2.1	No	2%	100%	MEDIUM	0%

Field	Description
GENE	Name of gene
POSITION	Genomic coordinates. An asterix indicates locally cis-phased with other reportable variants
HGVS	cDNA and protein HGVS
AF	Raw allelic frequency observed of allele
DEPTH	Total read depth of coverage of the variant site
VCN	Predicted number of copies of variant allele in each tumor cell (ie. adjusted for purity & ploidy)
CN	Predicted tumor copy number at variant location in each tumor cell (ie. adjusted for purity & ploidy)
HOTSPOT	Exact variant (chromosome, position, ref, alt) is a known pathogenic or likely pathogenic variant
BIALLELIC	Probability that the variant is biallelic (no wildtype present) based on predicted copy number & purity (see next page for details)
CLONAL	Probability that the variant is clonal (present in every tumor cell) based on predicted copy number & purity (see next page for details)
SOMATIC	Probability that the variant is somatic based on predicted copy number & purity (see next page for details)
RNA	Support and depth of coverage for variant in RNA (only shown if RNA analysed)
DRIVER	Predicted driver likelihood per gene based on algorithm trained on Hartwig WGS database and depending on mutation type, sample TMB, biallelic status (see next pages for details)

Somatic Variants – algorithm details

Somatic Likelihood

In TUMOR_ONLY mode, the likelihood that each variant is somatic is calculated by comparing the variant AF to the ”expected” VAFs for both germline and somatic variants given the sample purity & ploidy. For somatic variants, the expected VAFs are those that would imply the somatic VCN = {minor allele, major allele or <=1}. For germline variants the expected VAFs are deemed those that would imply the tumor VCN = 100% (homozygous) OR VCN = {minor allele or major allele}.

The minimum distance to each of the expected germline and somatic vaf sets is calculated and annotated as VAF_DIS_MIN in the vcf. The variant is assigned a ‘somatic likelihood’ based on these rules”

```
HIGH: abs(somatic_vaf_distance) - abs(germline_vaf_dist) < -0.05 OR IS HOTSPOT
LOW: abs(somatic_vaf_distance) - abs(germline_vaf_dist) > 0.08
MED: all other
```

Clonal Likelihood

The clonal likelihood is estimated by iteratively identifying peaks in the somatic variant copy number distribution.

For each peak we estimate a copy number, a total density and a distribution. Peaks with variant copy number < 0.85 copies are marked as subclonal.

For each variant, the likelihood that it belongs to a clonal vs subclonal peak is calculated

Biallelic Likelihood

The biallelic probability at a given variant position is calculated as the total probability between a loss of heterozygosity at the position, P(LOH), and the probability of the variant being present on all remaining alleles conditional on the LOH status, P(BA|LOH).

$$P(BA) = P(LOH) \times P(BA \mid LOH) + P(\text{not LOH}) \times P(BA \mid \text{not LOH})$$

Somatic Variants – algorithm details

Driver Likelihood

A driver likelihood estimate between 0 and 1 is calculated for each small variant in the gene panel. Hotspot mutations, biallelic nonsense and splice mutations for TSG and inframe mutations outside repeat contexts for oncogenes are automatically given a likelihood of 1.

The remaining point mutation variants are estimated based on the dN/dS (the rate of non-synonymous to synonymous mutations) in the Hartwig WGS pan-cancer cohort per variant type. Genes with an excess of non-synonymous mutation (based on calculated dNdS rates and after allowing for the already assigned drivers) will have a driver likelihood >0. Any remaining point mutations are assigned a driver likelihood between 0 and 1 using a bayesian statistic to calculate a sample specific likelihood of each gene based on the type of variant observed (missense, nonsense, splice or INDEL) and taking into account the mutational load of the sample.

The sample specific likelihood of a residual excess variant being a driver is estimated by:

$$P(\text{Driver}|\text{Variant}) = P(\text{Driver}) / (P(\text{Driver}) + P(\text{Variant}|\text{Non-Driver}) * (1-P(\text{Driver})))$$

Note that the algorithm is currently trained only on adult solid tumors & dN/dS would need to be retrained to be suitable for hematological or pediatric tumors

Where P(Driver) is assumed to be constant across samples. ie

$$P(\text{Driver}) = (\#\text{ residual unallocated drivers in gene}) / \#\text{ of samples in cohort}$$

And P(Variant|Non-Driver), the probability of observing n or more passenger variants of a particular variant type in a sample in a given gene, is assumed to vary according to tumor mutational burden, and is modelled as a poisson process:

$$P(\text{Variant}|\text{Non-Driver}) = 1 - \text{poisson}(\lambda = \text{TMB}(\text{Sample}) / \text{TMB}(\text{Cohort}) * (\#\text{ of passenger variants in cohort}), k = n-1)$$

Somatic Variants – Amplifications & Deletions

Amplifications and Deletions (1)

LOCATION	GENE	TYPE	RANGE	MIN CN	MAX CN	REL CN	ARM CN	DRIVER
chr10q23.31	PTEN	DEL	PARTIAL	0.0	2.0	0.0	2.0	HIGH

Field	Description
LOCATION	Chromosome & band of gene
GENE	HGNC Gene Name
TYPE	AMP (amplification), DEL (homozgous deletion), HET_DEL (heterozygous deletion) or LOH (Copy neutral loss of heterozygosity)
RANGE	Either FULL (full gene is impacted) or PARTIAL (partial gene is impacted)
MIN CN	Minimum exonic copy number across gene
MAX CN	Maximum exonic copy number across gene
REL CN	Relative copy number of the gene compared to tumor ploidy (= MIN_CN / PLOIDY)
ARM CN	Mean copy number of chromsome arm of gene. Can used to infer whether CN event is focal or arm level
DRIVER	Likelihood of driver. HIGH for DEL or AMP with REL_CN > threshold. LOW for HET_DEL, LOH and AMP with 85%*Threshold <REL_CN < Threshold

By default, LOH is only reported for MMRd & HRD associated genes

Amplification threshold is 3x ploidy by default but may be set in driver gene panel

Somatic Variants - Fusions

Fusions (4)

FUSION	JUNCTIONS	JCN	PHASING	TYPE	DRIVER
TMPRSS2::ERG	Exon 1 (ENST00000332149) - Exon 3 (ENST00000398897)	1.0	Inframe	KNOWN_PAIR	HIGH
BCL11A::FGFR1	Exon 2 (ENST00000642384) - Exon 3 (ENST00000447712)	4.1	Inframe	PROMISCUOUS_3	LOW
ETV6::FBXL14	Exon 1 (ENST00000396373) - Promoter Region (ENST00000339235)	0.1	Inframe	PROMISCUOUS_5	LOW
FGFR1::ANXA4	Exon 1 (ENST00000533668) - Exon 2 (ENST00000394295)	3.2	Inframe	PROMISCUOUS_5	LOW

Field	Description
FUSION	Fusion gene pair
JUNCTIONS	5' partner context and ensembl transcript and 3' partner context and transcript. Context may be 'Exon', 'Intron', 'Promoter', 'Upstream' or 'Downstream'
JCN	Junction copy number
PHASING	Either inframe (the fusion genes are transcribed inframe), out-of-frame or skipped exons
TYPE	For fusions which match the Hartwig fusion knowledgebase, one of KNOWN_PAIR (curated pathogenic gene fusions) , KNOWN_ENHANCER_PAIR (curated IG, TCR or BCLL1B rearrangements), or KNOWN_EXON_DEL_DUP (curated pathogenic deletions & duplication) PROMISCUOUS_3 & PROMISCUOUS_5 rearrangements involving curated promiscuous 3' and 5' partners are also shown
RNA FRAGS	Total fragment support for fusion in RNA (only shown if RNA is analysed)
DRIVER	HIGH if 'KNOWN' fusion type. LOW if 'PROMISCUOUS' type

Somatic Variants – Fusions (continued)

Essential features for a fusion to be reportable:

- 3’ gene partner must have coding bases
- 5’ gene partner transcript must have one of the following ensembl biotypes: ‘protein_coding’, ’retained_intron’, ’processed_transcript’, ’nonsense_mediated_decay’, ’lincRNA’
- The upstream breakend must fall within the 5’ partner transcript and be disruptive to the transcript
- The downstream breakend must fall either within the 3’ gene or within 100kb upstream
- The combined length of all segments in the chain must be less than 150kb
- The SV or chain must join appropriate contexts of the 5’ and 3’ genes and for coding regions must be inframe after allowing for any skipped exons. For exonic-to-exonic fusions exact base phasing is also checked as splice acceptor to splice donor phasing. The table to the right shows allowed contexts

	3' Fusion Partner Context					
5' Partner Context	Upstream	5'UTR Intronic	5'UTR Exonic	Coding Intronic	Coding Exonic	3' UTR or Non-Coding
Upstream	X (Enhancer)	X (promoter loss)				X (No Downstream Impact)
5'UTR or non-coding Intronic	YES (1)	YES	YES	alt start codon (4)		
5'UTR or non coding Exonic	exon skipped (3)		YES	exon skipped & alt start codon (3,4)	alt start codon (4)	
Coding Intronic	YES (1)	SEE NOTE (5)		YES	YES (2)	
Coding Exonic	exon skipped (3)				YES	
3'UTR	X (post stop codon in upstream gene)					

The notes in the table (1-5) and more details can be found here: https://github.com/hartwigmedical/hmftools/tree/master/linx#clustering-of-svs-into-events

Different reportability rules apply to enhancer rearrangements and exon deletion or duplications. Details available on github

Somatic Variants - Gene Disruptions

Disruptions (2)

GENE	POSITION	ZYGOSITY	CONTEXT	TYPE	JCN	UNDISRUPTED CN	DRIVER
PTEN	chr10:87940542 - chr10:87952584	HOM	Intron 5 - Intron 6	DEL	2.0	0.0	LOW

Field	Description
GENE	HGNC name of gene
POSITION	Exact breakpoints of breakends in the reference genome
ZYGOSITY	HET (heterozygous – has wildtype remaining) or HOM (homozygous – no wildtype remaining)
CONTEXT	Location of the start and end of disruption in the gene. If only one side of the junction impacts the gene only that context and the orientation of the breakpoint relative to the gene will be shown
TYPE	Structural variant type. One of {DEL, DUP, INV, BND, SGL or COMPLEX}
JCN	Junction copy number. Number of copies of the structural variant present in each tumor cell
UNDISRUPTED CN	Number of copies of the gene unaffected by this disruption in each tumor cell
DRIVER	Likelihood of being a driver. Set to LOW for heterozygous and high for homozygous disruptions

Partial gene deletions and amplification may also be reportable as disruptions

Somatic Variants - Viruses

Viruses (1)

VIRUS	QC STATUS	TYPE	INTEGRATIONS	% COVERED	MEAN COV	EXP CLONAL COV	DRIVER
Human papillomavirus 16	NO_ABNORMALITIES	HPV	9	97%	1772.6	15.8	HIGH

Field	Description
VIRUS	Detailed name of virus
QC STATUS	NO_ABNORMALITIES if qc status is fine. Otherwise one of {LOW_VIRAL_COVERAGE EXCESSIVE_VIRAL_COVERAGE, ASSEMBLY_DOWNSAMPLED, CHILD_TAXID_REFERENCE, UNCLEAR_TAXID_ASSIGNMENT}
TYPE	HET (heterozygous – has wildtype remaining) or HOM (homozygous – no wildtype remaining)
INTEGRATIONS	The number of detected integrations of this virus into the tumor genome (if 0, the virus is present but no integration site into the host DNA could be detected)
% COVERED	Percentage of the viral genome detected in the tumor sample
MEAN COV	Average coverage of the viral genome detected in the tumor sample
EXP CLONAL COV	The expected clonal coverage assuming the virus is clonally integrated only once in the tumor DNA
DRIVER	Likelihood of being a driver. Set to LOW for heterozygous and high for homozygous disruptions

Reportable Viruses
Human Papillomavirus (HPV)
Epstein-Barr virus (EBV)
Hepatitis B virus (HBV
Merkelcelpolyomacirus (MCV)
Humen herpesvirus (HHV-8)

Viruses are only detected in WGS based reports, not panels

Somatic Variants – Arm Copy Number

Arm Copy Number Abberations

CHROMOSOME	ARM	TYPE	CN	REL CN	DRIVER
chr1	Q	GAIN	3.9	1.4	HIGH
chrY	P	LOSS	0.0	0.0	HIGH
chrY	Q	LOSS	0.0	0.0	HIGH
chr1	P	LOSS	2.3	0.7	LOW
chr3	Q	GAIN	4.0	1.3	LOW
chr4	P	LOSS	2.0	0.7	LOW
chr4	Q	GAIN	3.8	1.3	LOW
chr5	P	LOSS	2.0	0.7	LOW
chr5	Q	LOSS	2.0	0.7	LOW

Field	Description
CHROMOSOME	Chromosome with gain or loss
ARM	Short (P) or long (Q) arm
TYPE	'GAIN' or 'LOSS'
CN	Average copy number of chromosome arm
REL CN	Average copy number of chromosome arm divided relative to ploidy of tumor. For male X & Y chromosomes, the CN is relative to the ploidy / 2
DRIVER	HIGH if GAIN is > 1.4x or LOSS is <0.6x REL CN. LOW if GAIN is > 1.25x or or LOSS < 0.75x REL CN

Somatic Variants - Signatures

Signatures are only included for WGS reports

Signature Allocations (12)

SIGNATURE	ETIOLOGY	ALLOCATION	PERCENT
Sig7	Ultraviolet light exposure	23623	60%
Sig2	APOBEC activity	3862	10%
Sig8	Unknown	2312	6%
Sig18	Reactive oxygen species	2309	6%
Sig4	Tobacco smoking	2236	6%
Sig12	Unknown	1644	4%
Sig11	Temozolomide treatment	1529	4%
Sig17	Unknown	1012	3%
Sig28	Unknown	562	1%
Sig24	Aflatoxin exposure	314	1%
Sig3	Defective HR	259	1%
MISALLOC	-	4984	13%

MISALLOC counts the # of variants that could not be assigned to a known signature

Field	Description
SIGNATURE	Cosmic v2 SBS signature (https://cancer.sanger.ac.uk/signatures/signatures_v2/)
ETIOLOGY	Proposed etiology of SBS signature
ALLOCATION	# of somatic variants allocated to the signature
PERCENT	% of somatic variants allocated to signature

Rule of Thumb: A signature may be significant if it contains either >20% or >2000 somatic variants

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Germline Variants

Germline Variants – Small Variants

Small variants (1)

VARIANT	HGVS	ZYGOSITY	AF	DEPTH	VCN	CN	MACN	HOTSPOT	BIALLELIC	GNOMAD	CLINVAR	DRIVER
chr13:32330989 CAG>C	p.Asp252fs	HET	0.57	61	0.7	1.0	0.0	Yes	Yes	0.00e+00	PATHOGENIC	HIGH

Field	Description
GENE	Name of gene
VARIANT	Genomic coordinates. An asterisk indicates locally cis-phased with other reportable variants
HGVS	cDNA and protein HGVS
ZYGOSITY	Genotype in the normal sample HET or HOM
AF	Raw allelic frequency observed of allele in the tumor sample
DEPTH	Total depth of coverage of the variant site in the tumor sample
COPIES	Predicted number of copies of variant allele in each tumor cell and predicted tumor copy number at variant location in each tumor cell
HOTSPOT	Exact variant (chromosome, position, ref, alt) is a known pathogenic or likely pathogenic germline variant
BIALLELIC	Probability that the variant is biallelic (no wildtype present) in the tumor based on predicted copy number & purity
GNOMAD	GNOMAD population frequency of variant
CLINVAR	CLINVAR pathogenic status for variants which match clinvar
DRIVER	Set to HIGH for pathogenic germline small variants. Other variants are not reported

Germline Variants – Amplifications & Deletions

Amplifications and Deletions (1)

LOCATION	GENE	TYPE	RANGE	MIN CN	MAX CN	REL CN	ARM CN	DRIVER
chr19p13.3	STK11	HET	Exon 9 - Exon 10	0.3	2.8	0.3	2.9	HIGH

Field	Description
LOCATION	Chromosome arm and band
GENE	HGNC gene name
TYPE	Zygosity in germline – either HET (heterozygous) or HOM (homozygous)
RANGE	Either 'FULL' gene or exon range affected
MIN CN	Minimum exonic copy number across gene in tumor cells
MAX CN	Maximum exonic copy number across gene in tumor cells
REL CN	MIN CN divided by ploidy of tumor
ARM CN	Average copy number of arm of gene
DRIVER	HIGH or LOW (currently always set to high for germline amplifications and deletions)

Germline Variants – Disruptions

Disruptions (1)

GENE	POSITION	ZYGOSITY	CONTEXT	TYPE	JCN	UNDISRUPTED CN	DRIVER
BRCA1	chr17:43117573	HOM	Intron 2 Downstream	DEL	2.0	0.5	HIGH

Field	Description
GENE	HGNC gene name
POSITION	Chromosome and position of the disruptive breakend(s)
ZYGOSITY	Zygosity of the disruption in the germline sample
CONTEXT	Location of the start and end of disruption in the gene. If only one side of the junction impacts the gene only that context and the orientation of the breakpoint relative to the gene will be shown
TYPE	Structural variant type. One of {DEL, DUP, INV, BND, SGL or COMPLEX}
JCN	Junction copy number. Number of copies of structural variant present in each tumor cell
UNDISRUPTED CN	Number of copies of the gene unaffected by this disruption in each tumor cell
DRIVER	Likelihood of being a driver. Set to LOW for heterozygous and high for homozygous disruptions

Germline Variants – chromosomal aberrations

Chromosomal aberrations (2)

MOSAIC_X, TRISOMY_13

Identifies any known chromosomal aberrations in the germline sample if provided. Supported aberrations:

Aberration	Chromosome affected
MOSAIC_X	X
KLINEFELTER	X
TRISOMY_<CHR>	X, 21,13,18,15, 9P
TETRASOMY_9P	9P

Germline Variants - Pharmacogenetics

Pharmacogenetics (1)

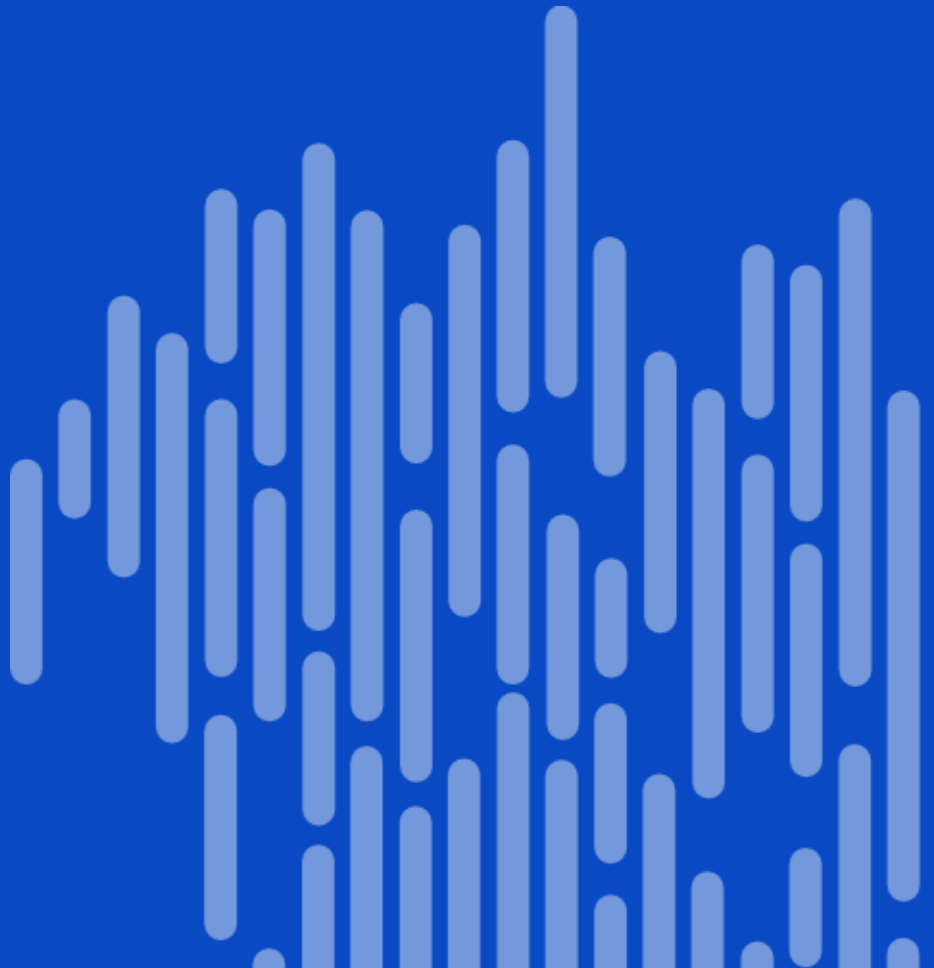
GENE	HAPLOTYPE	GENOTYPE	FUNCTION	LINKED DRUGS	SOURCE
DPYD	*1	HOM	Normal Function	5-Fluorouracil;Capecitabine;Tegafur	PHARMGKB

Only DPYD is currently supported

Field	Description
GENE	HGNC gene name
HAPLOTYPE	Specific haplotype allele found in germline sample
GENOTYPE	Zygosity of haplotype in germline – either HET (heterozygous) or HOM (homozygous)
FUNCTION	Efficacy of the respective enzymes in the haplotype
LINKED DRUGS	Specific drugs linked to the efficacy of the respective enzyme.
SOURCE	Source of the efficacy data

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RNA



RNA – QC

RNA Findings

QC

RNA sample is optional in oncoanalyser

QC	TOTAL FRAGMENTS	DUPLICATE RATE	SPLICED RATE	UNSPLICED RATE	ALT-SLICED RATE	CHIMERIC RATE
PASS	11832887	80%	85%	7%	5%	2%

Field	Description
QC	QC status of RNA sample
TOTAL FRAGMENTS	Total fragments in RNA sample
DUPLICATE RATE	Rate of duplicates in RNA sample
SPLICED RATE	Proportion of genic fragments supporting normal 'spliced' transcripts
UNSPLICED RATE	Proportion of genic fragments supporting non-splice transcripts. ie fragments partially or fully support intronic regions of the gene and may reflect pre-spliced RNA transcripts
ALT-SPLICED RATE	Proportion of genic fragments supporting alternate 'spliced' transcripts. ie supporting novel splicing but within 500kb of the gene with both splice sites on the same strand
CHIMERIC RATE	Proportion of genic fragments supporting chimerically spliced transcripts. ie with other splice site either inverted, inter-chromosomal or more than 500kb from the gene

RNA QC statistics currently reported separately to main QC report which is DNA only

RNA – High Gene Expression

High Gene Expression (1)

GENE	TPM	PERCENTILE	FOLD CHANGE
EGFR	595.9	0.98	17.8

Field	Description
GENE	HGNC gene name
TPM	Estimated transcript per million for gene expression
PERCENTILE	TPM percentile compared to Hartwig pan-cancer WTS cohort
FOLD CHANGE	TPM divided by median Hartwig pan-cancer WTS cohort

High expression is reported if both TPM is greater than 98th percentile pan-cancer and log2(fold change) > 1 OR if log2(fold change) > 2.3

RNA – Fusions detected in RNA, not DNA

Fusions Detected In RNA, Not In DNA (4)

FUSION	TYPE	SV TYPE	JUNC START	JUNC END	SUPPORT	COHORT FREQ
PAX8::PPARG	KNOWN_PAIR	BND	KNOWN	KNOWN	5/26	0
TFG::NTRK1	KNOWN_PAIR	BND	KNOWN	KNOWN	3/709	0
ETV6::NTRK3	KNOWN_PAIR	BND	KNOWN	KNOWN	13/47	0
TMPRSS2::ERG	KNOWN_PAIR	DEL	KNOWN	KNOWN	1/1	11

Field	Description
FUSION	Gene names of fused genes <5' partner>::<3' partner>
TYPE	KNOWN (present in fusion KB) or PROMISCUOUS (5' or 3' partner matches a promiscuous fusion gene)
SV TYPE	Type of structural variant causing the fusion (BND= translocation)
JUNC START	KNOWN (matches known exon boundary), CANONICAL (matches canonical splice sequence) or NONE (other)
JUNC END	KNOWN (matches known exon boundary), CANONICAL (matches canonical splice sequence) or NONE (other)
SUPPORT	# of fragments supporting and total depth at junction boundary
COHORT FREQ	# of observations in Hartwig WTS cohort

Only additional fusions NOT found in DNA are shown in this section

RNA – Novel Splice Junctions

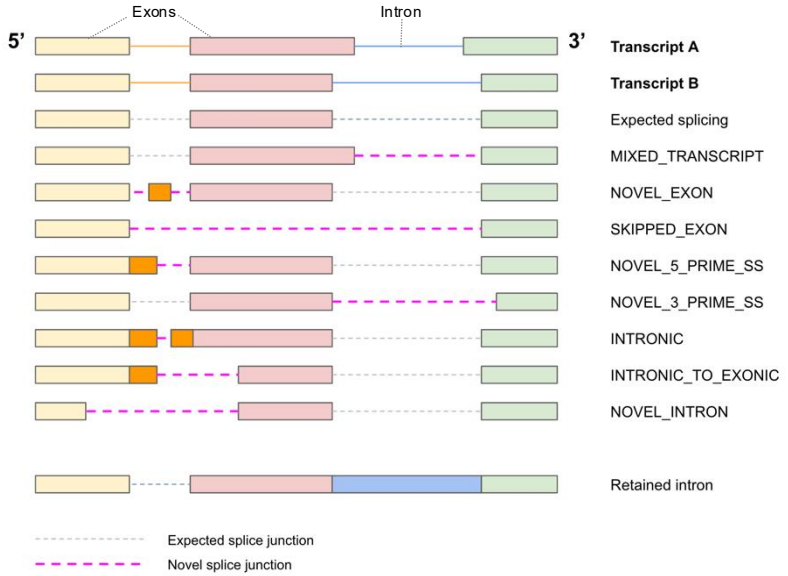
Novel Splice Junctions (1)

GENE	JUNCTIONS	TYPE	JUNC START	JUNC END	SUPPORT	COHORT FREQ
EGFR	Exon 1 (7:55087058) - Exon 8 (7:55223523)	SKIPPED_EXONS	SPLICE_JUNC	SPLICE_JUNC	3723/8512	15

Field	Description
GENE	HGNC gene name
JUNCTIONS	Start and end exon and position
TYPE	Type of novel splicing. See diagram
JUNC START	Gene context at start junction position. One of 'SPLICE_JUNC','EXONIC' or 'INTRONIC'
JUNC END	Gene context at start junction position. One of 'SPLICE_JUNC','EXONIC' or 'INTRONIC'
SUPPORT	# of fragments support and depth at splice location
COHORT FREQ	# of observations in Hartwig WTS cohort

Novel splice junctions are reportable if they match the gene and range of curated EXON_DEL_DUP in the fusion knowledgebase

Effects of splicing on transcript structure



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Immunology

Immunology – HLA class I and II

Human leukocyte antigen (HLA) Class I proteins present antigens on the cell surface (HLA-A, HLA-B and HLA-C)

HLA Class I Alleles

ALLELE	QC STATUS	REF FRAGS	TUMOR FRAGS	TUMOR CN	SOMATIC MUTATIONS
A*01:01	PASS	258	706	2.0	NONE
A*01:01	PASS	258	707	1.4	NONE
B*08:01	PASS	216	640	1.4	NONE
B*40:02	PASS	212	751	2.0	NONE
C*03:04	PASS	249	803	2.0	NONE
C*07:01	PASS	225	621	1.4	NONE

HLA Class II Alleles

ALLELE	QC STATUS	REF FRAGS	TUMOR FRAGS	TUMOR CN	SOMATIC MUTATIONS
DPA1*01:03	PASS	127	440	2.0	NONE
DPA1*02:01	PASS	103	296	1.4	NONE
DPB1*01:01	PASS	150	374	1.4	NONE
DPB1*02:01	PASS	129	491	2.0	NONE
DQA1*05:01	PASS	92	293	2.0	NONE
DQA1*05:05	PASS	99	298	1.4	NONE
DQB1*02:01	PASS	106	314	1.4	NONE
DQB1*03:01	PASS	113	415	2.0	NONE

Field	Description
ALLELE	The specific IMGT alleles found in the patient sample
QC STATUS	Either PASS or WARN (for low coverage OR high support for unexpected haplotypes OR indels)
REF FRAGS	# of supporting fragments for allele in germline sample
TUMOR FRAGS	# of supporting fragments for allele in tumor sample
RNA FRAGS	# of supporting fragments for allele in RNA sample if RNA data is provided
TUMOR CN	Predicated number of copies of the allele in the tumor sample
SOMATIC MUTATIONS	Somatic small variant mutations found to be present in this allele

HLA Class II present peptides on professional antigen presenting cells. Only DPA1, DPB1, DQA1 & DQB1 are evaluated

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Tissue of Origin

Cancer of Unknown Primary Prediction Algorithm (CUPPA)

CUPPA (PMID: 36463731) predicts the tissue of origin using features derived from whole genome sequencing (WGS; DNA). The algorithm is a collection of per-feature-type logistic regressions:

DNA classifiers

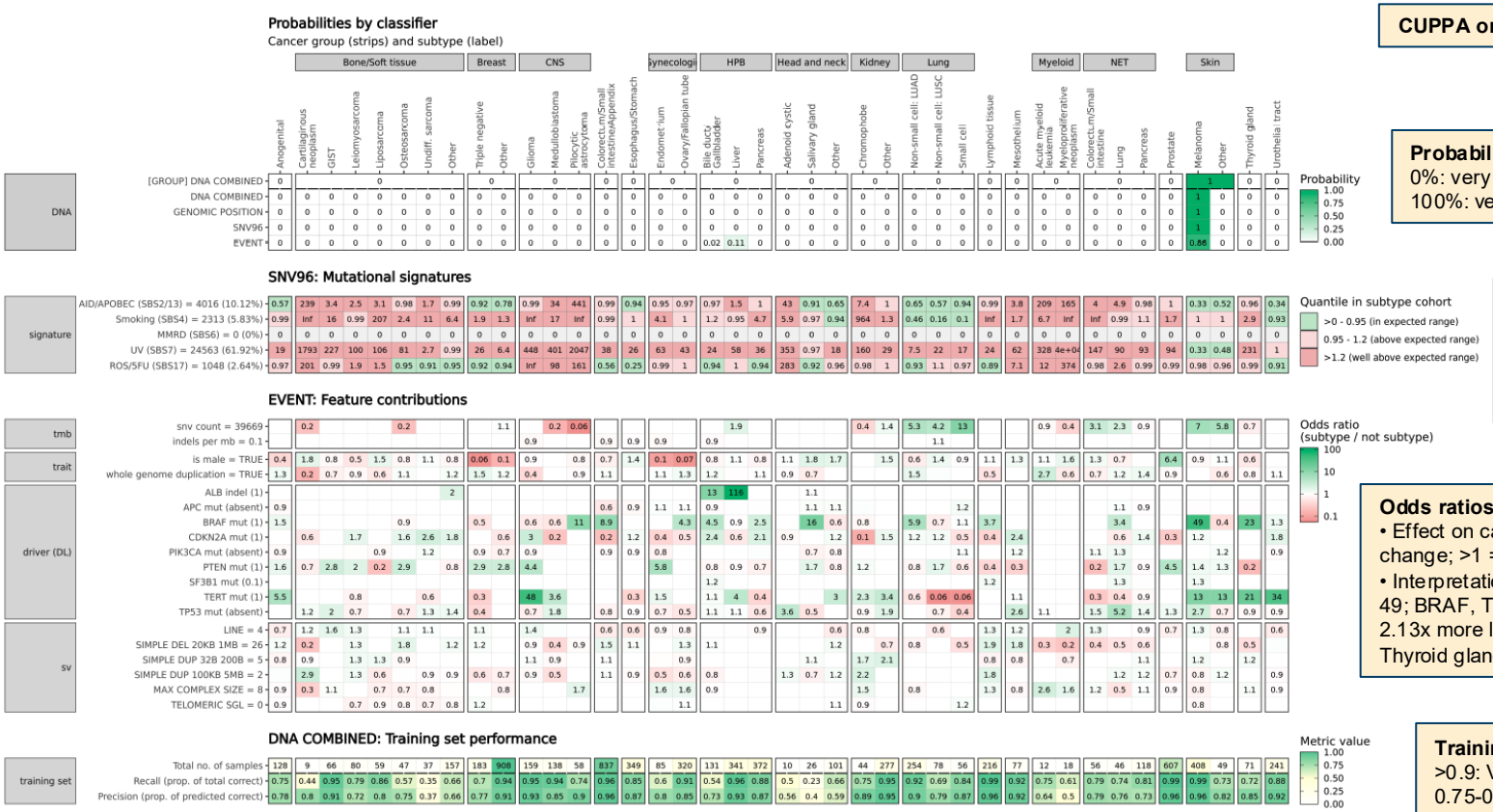
- **GEN_POS**: Uses SNV counts at each genomic position (500kb bins) to detect patterns that are associated with cancer-type-specific epigenetic state
- **SNV96**: Uses 96-trinucleotide context (SNV) counts to detect mutational patterns (e.g. UV radiation or smoking). The classifier does not use COSMIC signatures directly, but signature contribution percentiles (relative to the Hartwig cohort) are shown for informational purposes in the CUPPA report
- **EVENT**: Uses various cancer-type-specific mutational “events” (e.g. driver mutations, fusions, viral insertions, etc)

RNA classifiers

- **GENE_EXP**: Uses cancer-type-specific gene expression profiles
- **ALT_SJ**: Uses alternative splice junction profiles (i.e. those not annotated in ENSEMBL)

Note: TOO prediction for panels in oncoanalyser is based off a separate classifier vCUPPA and is not currently available in the Orange report

Cancer of Unknown Primary Prediction Algorithm (CUPPA)



CUPPA only runs on WG(T)S

Probabilities
0%: very unlikely
100%: very likely

Quantiles
if less than 100% gives quantile in cohort. For >100% is the multiple of highest cohort value.

Odds ratios (event classifier)
- Effect on cancer type probability: 1 = No change; >1 = Increase; <1 = Decrease.
- Interpretation: BRAF, Melanoma = 49; BRAF, Thyroid gland = 23; 49 / 23 = 2.13x more likely to be Melanoma than Thyroid gland cancer

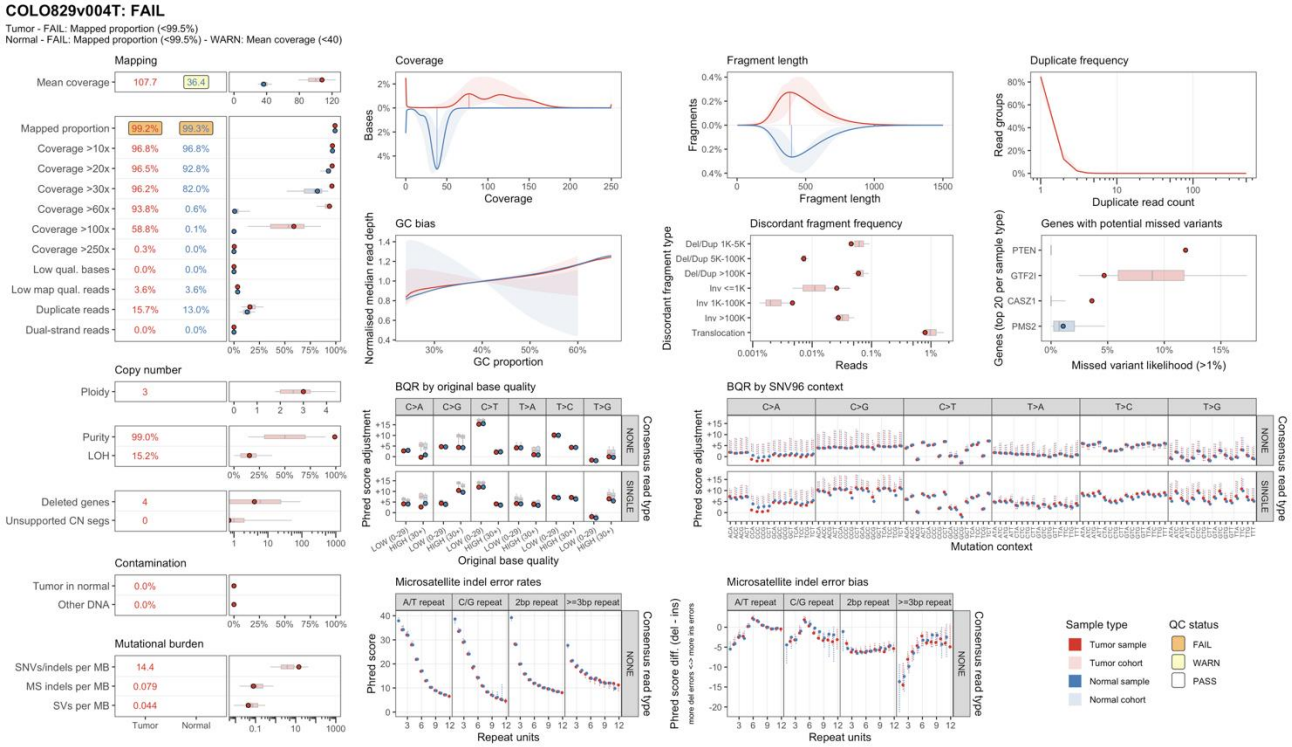
Training set performance
>0.9: Very reliable
0.75-0.9: Moderately reliable
<0.75: Potentially unreliable

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Quality Control

Quality Control (QC) – Report Overview

The QC section is a single page summary of all DNA QC statistics for the sample computed in oncoanalyser



Tumor sample statistics are shown in red. Normal sample in blue.

Cohort percentiles are shown as:

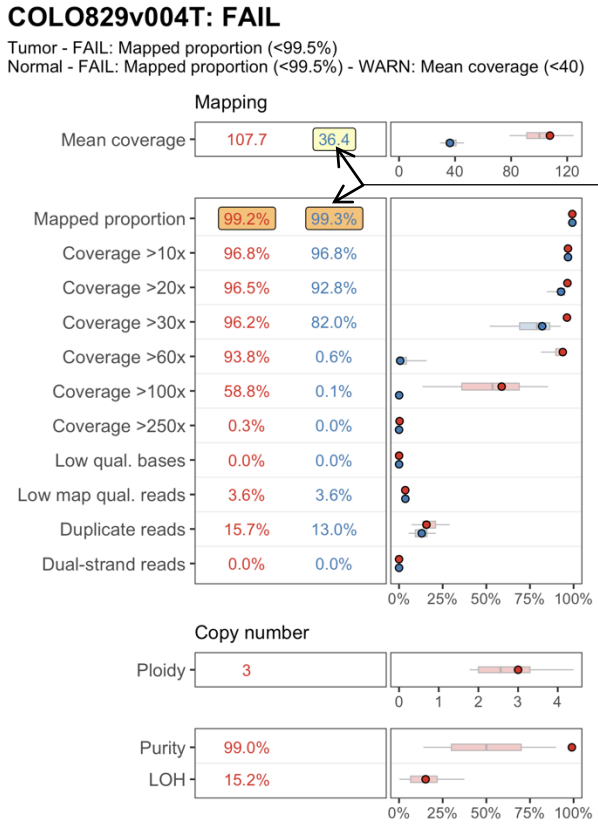
- Boxplots: 5th, 25th (Q1), 50th (median), 75th (Q3), and 95th percentiles
- Ribbons or vertical dotted lines: 5th and 95th percentiles

For custom panels, cohort data can be generated if >=50 samples are provided

Note: RNA QC is currently shown separately in the RNA section of the report

Note: Thresholds for "Mean coverage" and "Mapped proportion" were artificially increased to demonstrate WARN/FAIL annotations.

QC – Summary metrics



← Overall status: FAIL if any FAIL, else WARN if any WARN, else PASS

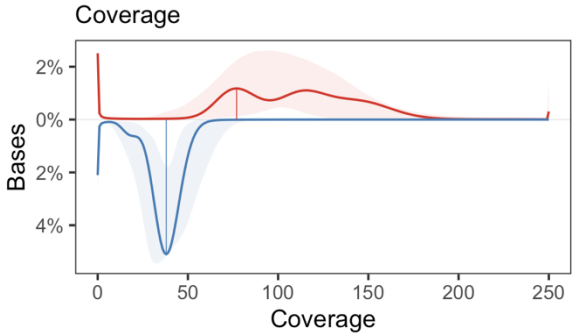
Numbers are highlighted if value does not meet FAIL or WARN threshold

QC status
- FAIL
- WARN
- PASS

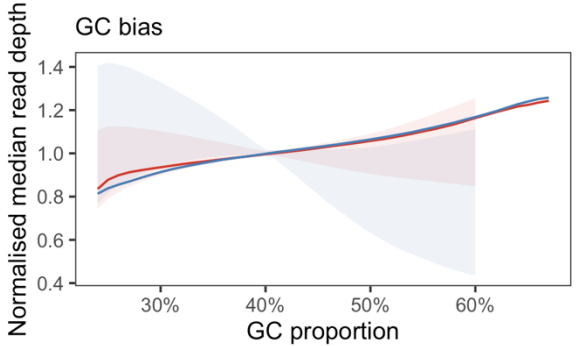
- Dot/Number: Value in tumor sample
- Boxplot: Distribution in tumor samples in Hartwig cohort

If normal sample provided:
- Dot/Number: Value in normal sample
- Boxplot: Distribution in normal samples in Hartwig cohort

QC – Plots

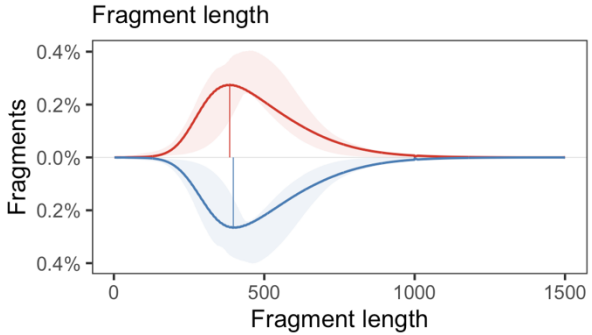


% of bases in genome at each coverage level

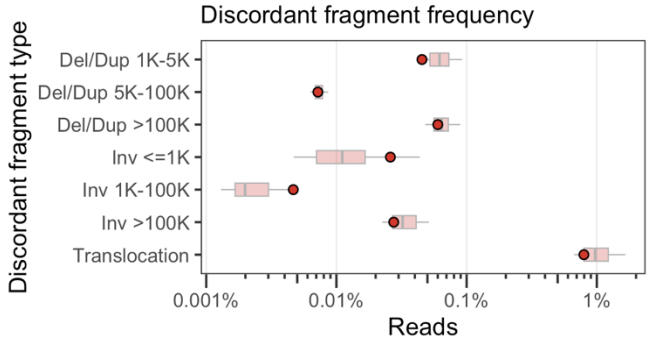


For each GC% bin, the median read depth across all windows determined to have a particular GC%, normalised (i.e. divided) by the global median read depth.

QC – Plots (continued)



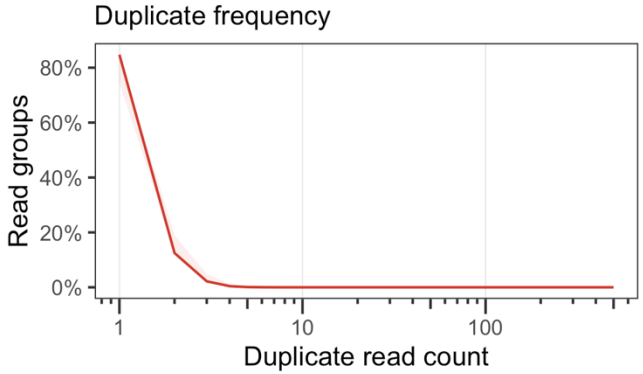
% of total fragments per fragment length bin



% of total reads per discordant fragment type

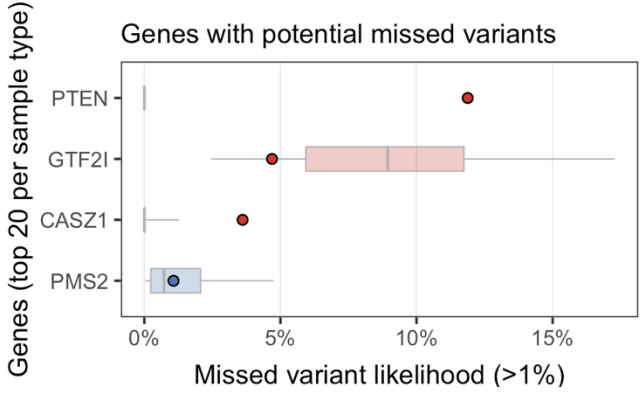
Discordant fragment type refers to the orientation/spacing of the paired end reads, and does not necessarily represent a structural variant

QC – Plots (continued)



% of read groups containing N duplicate reads

N=1 : Read groups without duplicate reads
N>=2: Read groups with duplicate reads

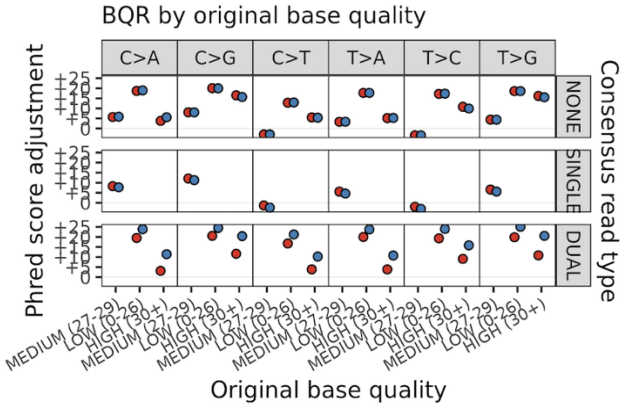


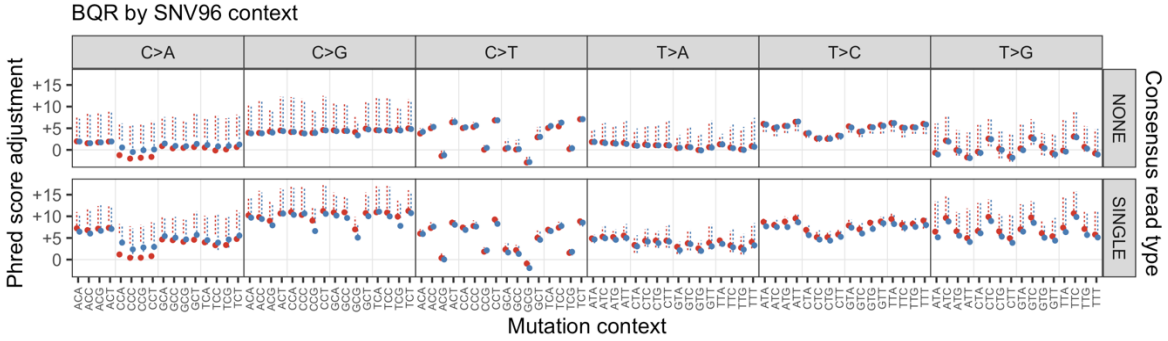
Likelihood that a variant is missed within the respective gene. Only genes (up to 20) with missed variant likelihood >1% are shown

QC – Plots (continued)

Base quality recalibration (BQR) of variants to account for sequencer-specific base quality biases:
+ : Base quality boosted (original quality underestimated)
− : Base quality reduced (original quality overestimated)

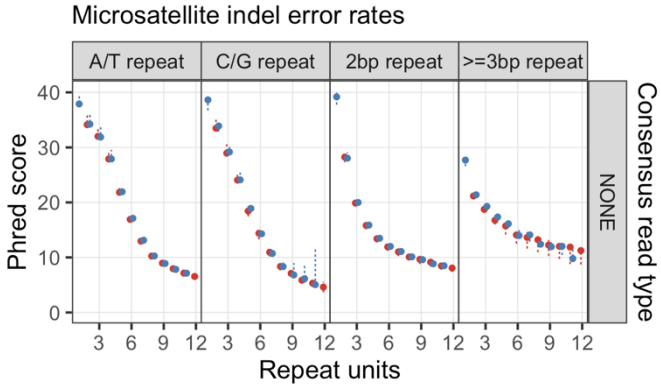
Consensus read types:
- NONE: No read deduplication (unique read)
- SINGLE: Read deduplication based on one strand (forward or reverse)
- DUAL: Read deduplication based on both forward and reverse strands



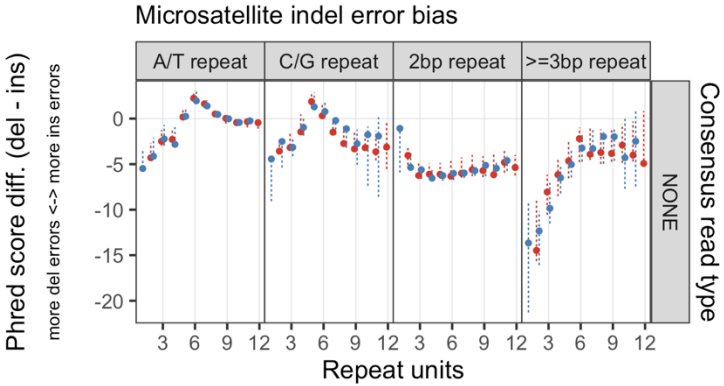


QC – Plots (continued)

Variation in repeat units (“error”) amongst reads (per consensus type; see previous slide) mapping to sites in the reference genome with a particular repeat type



Whether reads have more insertion or deletion errors per reference genome repeat type

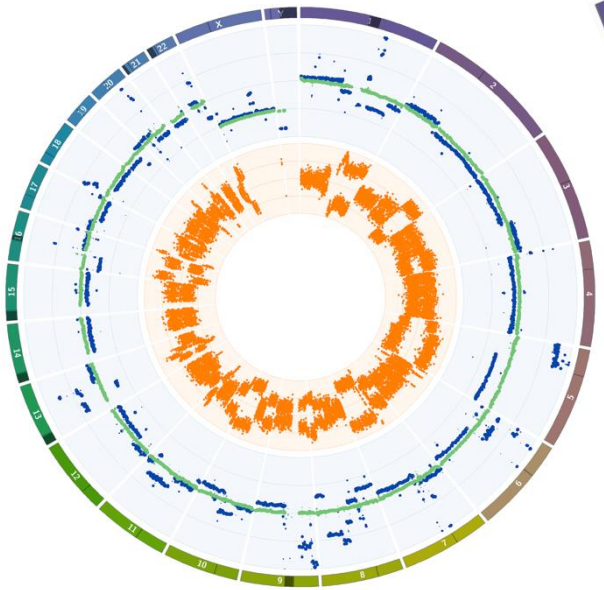


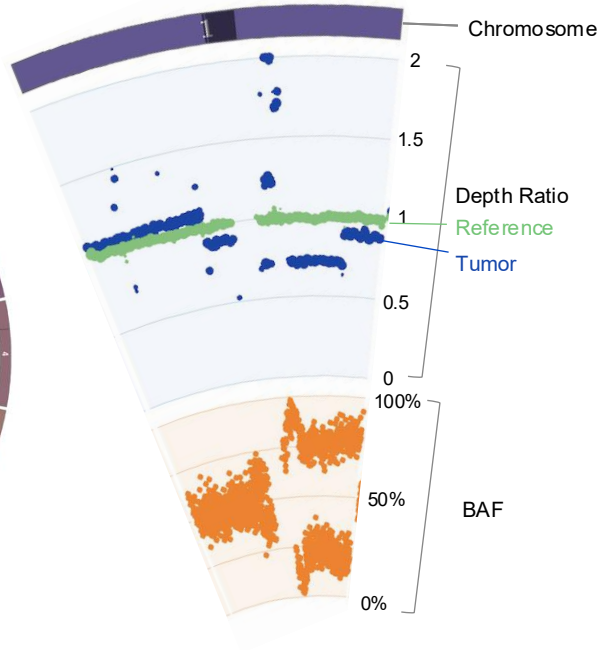
Hartwig Medical Foundation

Purity & Ploidy Plots

Purity & Ploidy – Copy Number Input CIRCOS

WGS input CIRCOS





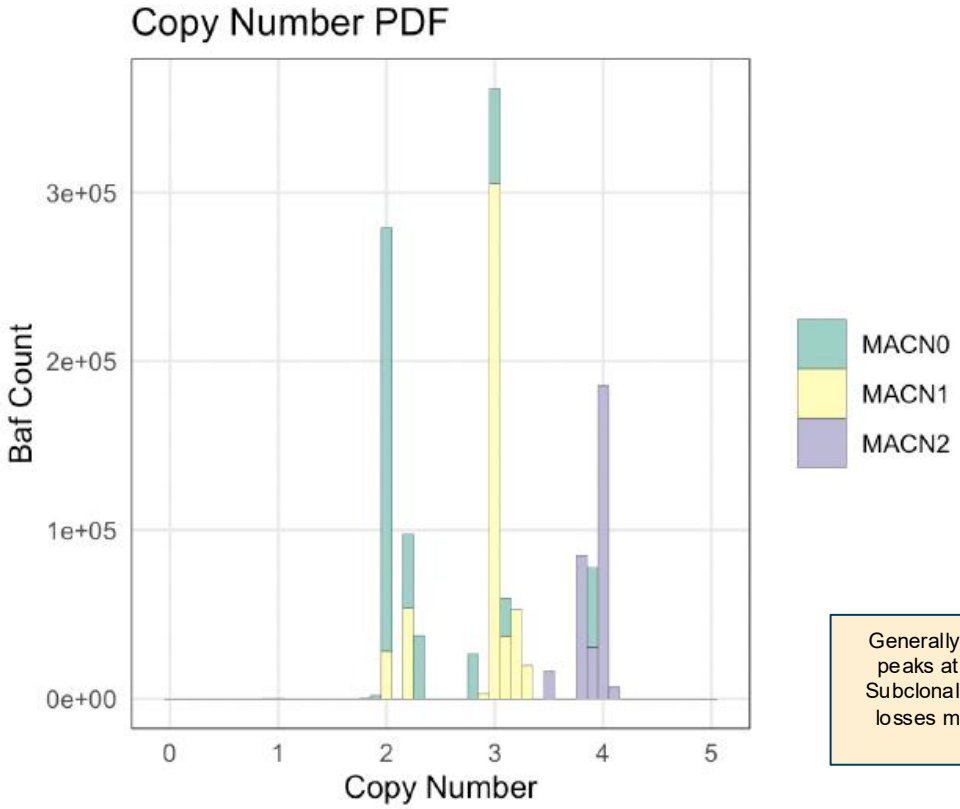
In tumor only, the green reference ratio is not shown

The input CIRCOS shows the main copy number inputs for the fit

Depth ratios of the reference (green) and tumor (blue) samples. The ratio is defined as the GC and expected enrichment (for targeted) normalised depth divided by the mean depth for the sample. For the reference, diploid normalization has also been applied. Only regions which are high confidence diploid in the germline are shown.

Observed B-Allele Frequency (BAF) of each germline polymorphism by chromosome and position. The number of points are down sampled to a maximum of 25k.

Purity & Ploidy – Copy Number Probability Density Function (PDF)

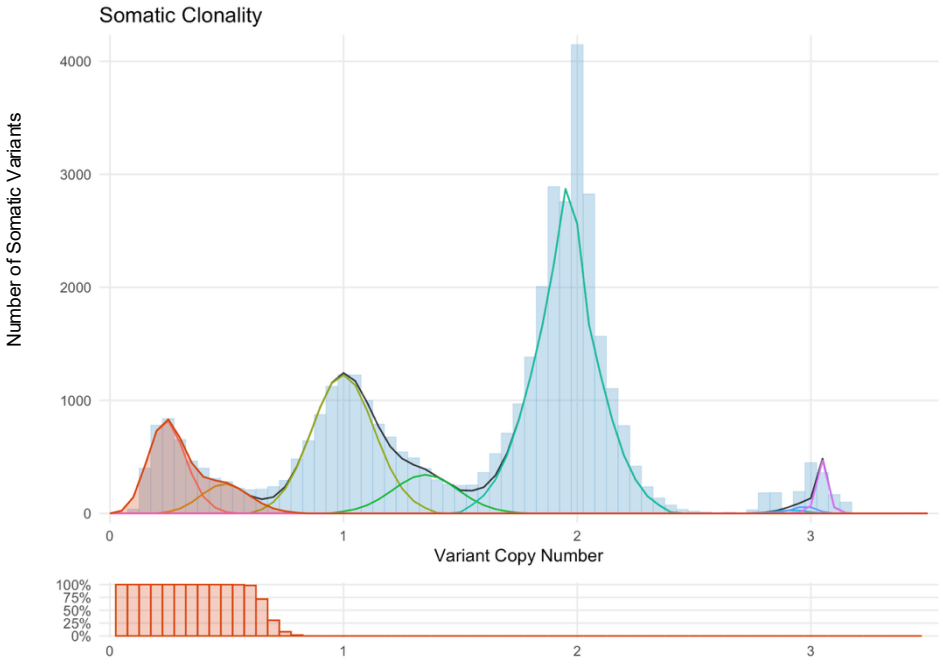


The copy number PDF shows the distribution of copy number states observed, weighted by the number of B-Allele Frequency (BAF) points in those segments.

Copy numbers are sliced using colour into their respective minor allele copy number (MACN).

Generally, we expect to see clear peaks at integer copy numbers. Subclonal copy number gains and losses may occur at non integer values

Purity & Ploidy – Somatic Clonality

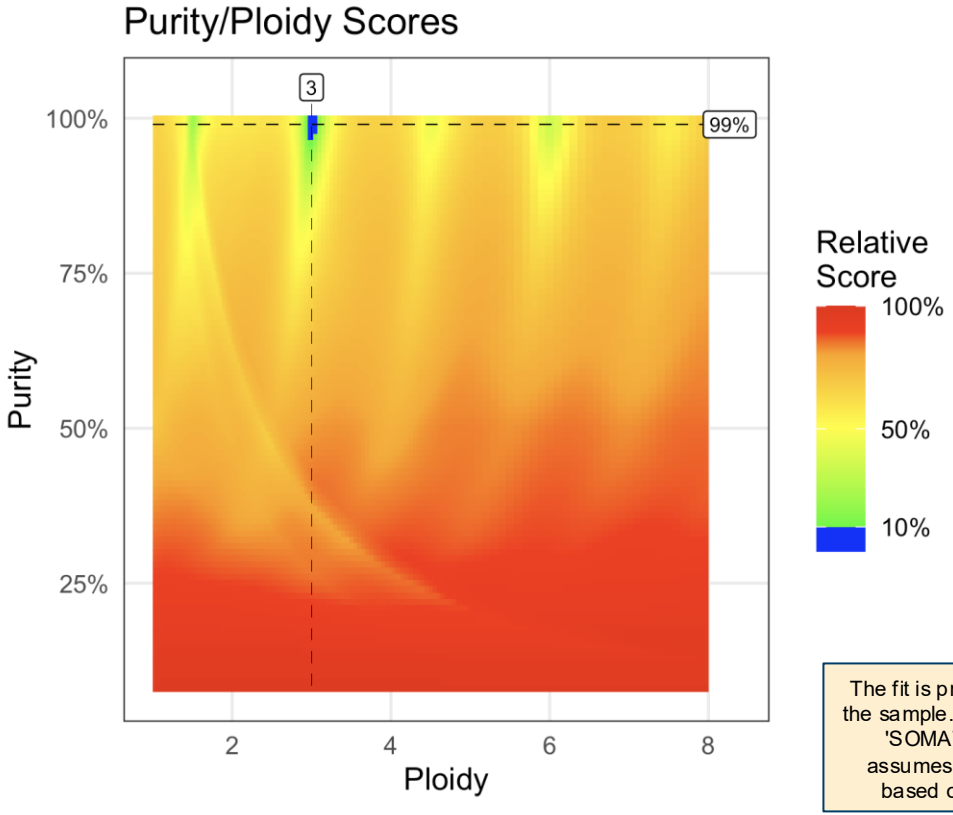


This plot shows the evaluation of clonality of all somatic SNV.

In the top histogram, the blue bars show the number of somatic variants (y-axis) for each variant copy number (X-axis). The coloured lines show the distribution of fitted variant peaks with red shaded peaks predicted to correspond to subclonal vaiants (ie VCN < 1)

The bottom chart predicts the likelihood of a variant being subclonal at any given variant copy number based on the fitted peaks in the top chart

Purity & Ploidy – Sunrise Plot

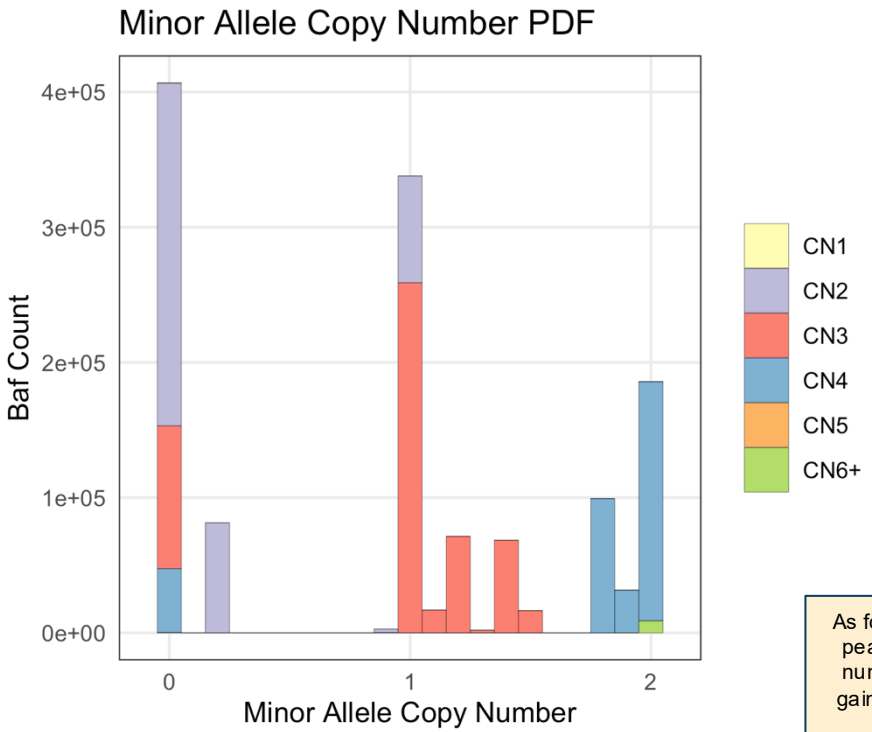


The sunrise plot shows the relative scoring for the full matrix of purities and ploidy solutions tested to determine the purity and ploidy fit.

The best fitting is marked. Solutions that score with 10% of the best score are coloured blue and are used to determine the uncertainty range for both the final purity and ploidy.

The fit is primarily based off the aneuploidy in the sample. For samples with low aneuploidy a 'SOMATIC' fit mode is triggered which assumes ploidy = 2 and determines purity based off AF of somatic small variants.

Purity & Ploidy – Minor Allele Copy Number PDF



The minor allele number PDF shows the distribution of minor allele copy numbers observed, weighted by the number of BAF points in those segments.

Copy numbers are sliced using colour into their respective copy number.

As for the CN PDF, we expect clear peaks at integer minor allele copy numbers. Subclonal copy number gains and losses may occur at non integer values

Purity & Ploidy - Kataegis plot

A Kataegis plot identifies and displays regions of localized hypermutation. The plot shows the distance between consecutive somatic base pair mutations (intermutation distance) on the y-axis (log10 scale), versus the genomic position (x-axis).

Somatic variants of type C>T and C>G in a TpCpN context are annotated as showing Kataegis (Nik Zainal et al., 2012) if there are three or more mutations of that type, strand and context localised within a region with an average inter-mutation distance of <= 1kb. The annotation describes the strand it was found on along with an id and is shared by grouped variants.

